Toulouse School of Economics

Time Series (M2 Economics of Global Risks)

Time Series Homework

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1 Problem I

Data

For Problem I, we use United States macroeconomic and energy data. The assignment requires a price index, an inflation series, GDP, a short-term interest rate, and daily electricity consumption for a developed country. The first four series are collected from the Federal Reserve Bank of St. Louis database (FRED); the electricity series is obtained from the U.S. Energy Information Administration (EIA). Because the electricity data begin only in January 2019, the **strict common calendar span** across all five variables is **January 2019–December 2025**. When the variables are studied individually, however, the three FRED monthly series and the quarterly GDP series are retained from **July 2015 onward**, providing a longer estimation window of roughly ten and a half years.

Price index. The price index is the *Consumer Price Index for All Urban Consumers: All Items in U.S. City Average* (`CPIAUCSL`) Federal Reserve Bank of St. Louis (n.d.a). The series is observed at a **monthly** frequency and is seasonally adjusted by the Bureau of Labor Statistics. We retain the sample from **July 2015 to December 2025** (126 monthly observations). Following the instructions of the exercise, we work with the logarithm of the index, $\log(\text{CPI}_t)$.

Inflation. Inflation is constructed from `CPIAUCSL` as the monthly log-difference of the price index, scaled to percentage points:

$$\pi_t \;=\; 100\left[\ln(\text{CPI}_t) - \ln(\text{CPI}_{t-1})\right].$$

Because the formula requires one lag of the CPI, one observation is lost and the inflation series runs from **August 2015 to December 2025** (125 monthly observations).

Gross Domestic Product. GDP is measured with the FRED series `GDP` Federal Reserve Bank of St. Louis (n.d.e). This variable is available at a **quarterly** frequency and is reported in billions of chained dollars at a seasonally adjusted annual rate. The raw download covers 2015Q1 onward; we retain the full download so that the sample runs from **2015Q1 to 2025Q4** (44 quarterly observations). As suggested in the assignment, we work with the logarithm of GDP, $\log(\text{GDP}_t)$.

Short-term interest rate. The short-term interest rate is proxied by the *Federal Funds Effective Rate* (`FEDFUNDS`) Federal Reserve Bank of St. Louis (n.d.c). This series is observed at a **monthly** frequency, is reported in percent per annum, and is not seasonally adjusted. We retain the sample from **July 2015 to December 2025** (126 monthly observations).

Daily electricity demand. Daily electricity consumption is proxied by *daily electricity demand* from the EIA open-data endpoint, based on Form EIA-930 and the Hourly Electric Grid Monitor U.S. Energy Information Administration (n.d.). We use the `US48` respondent, which aggregates demand across the contiguous United States (Lower 48 states). The variable is observed at a **daily** frequency and is measured in megawatt-hours (MWh). The downloaded files cover **1 January 2019 to 31 December 2025** (2,557 daily observations before any cleaning). The raw EIA output reports each daily aggregate in five time-zone versions (Eastern, Central, Mountain, Arizona, and Pacific); we retain only the `type = D` (Demand) observations for the **Eastern** timezone and discard all duplicates, yielding a unique daily series.

Summary of frequencies and samples. Table 1 collects the key features of each series.

Table 1: Data summary for Problem I

Series	Source	Frequency	Sample	T
$\log(\text{CPI}_t)$	FRED `CPIAUCSL`	Monthly	Jul 2015 – Dec 2025	126
π_t (inflation)	derived from CPI	Monthly	Aug 2015 – Dec 2025	125
$\log(\text{GDP}_t)$	FRED `GDP`	Quarterly	2015Q1 – 2025Q4	44
r_t (fed funds rate)	FRED `FEDFUNDS`	Monthly	Jul 2015 – Dec 2025	126
Electricity demand	EIA US48	Daily	Jan 2019 – Dec 2025	2,557

1.1 Data construction

All data used in Problem I are loaded and processed in a single R script. The script relies on the `tidyverse`, `jsonlite`, and `lubridate` packages.

The three macroeconomic series: CPI, GDP, and the federal funds rate are read directly from the FRED `.csv` downloads. For the CPI, one observation (October 2025) was missing from the download at the time of access and is dropped. Inflation is then computed as the monthly log-difference of the price index, scaled to percentage points, so the inflation series starts one period later than the CPI in August 2015. The CPI and the federal funds rate are merged into a single monthly dataframe by date. GDP is kept in a separate quarterly dataframe, since mixing frequencies in a single rectangular object would require aggregation or interpolation that is not warranted here.

The electricity series is assembled from 18 semi-annual JSON files downloaded from the EIA open-data API. Each file is parsed, filtered to keep only demand observations (`type = D`) in the Eastern timezone, and the 18 resulting dataframes are stacked. A final check verifies that the daily series is complete with no gaps between January 2019 and December 2025.

The script therefore produces three output files at different frequencies: `monthly_data.csv`, `quarterly_data.csv`, and `daily_data.csv`, which are loaded separately in the analysis that follows.

1.2 Time series properties: plot, descriptive statistics, ACF and PACF

Log CPI

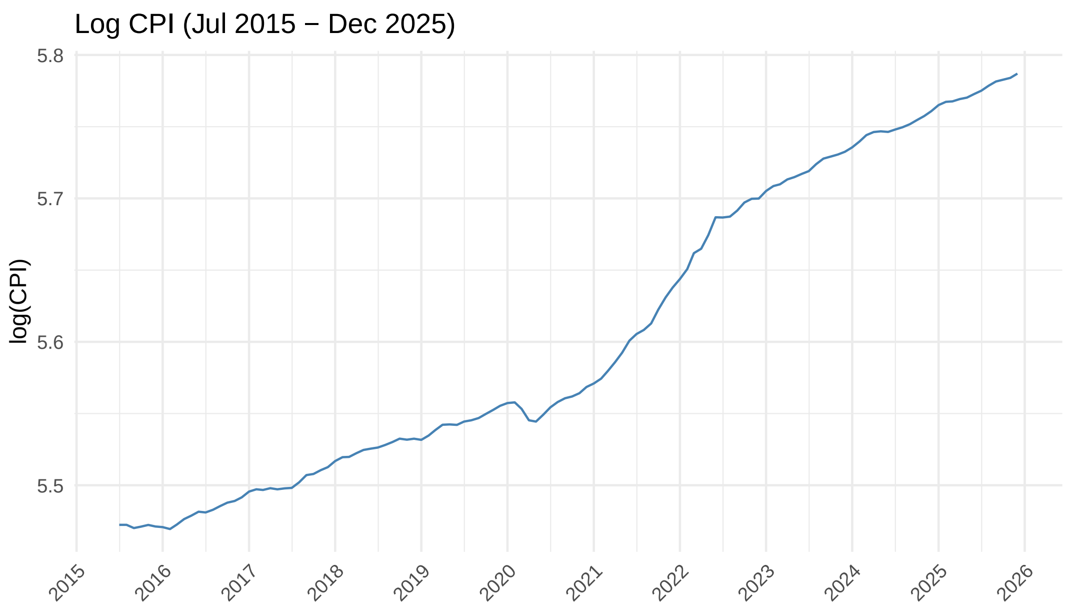

Figure 1: Log CPI, July 2015 – December 2025.

Figure 1 plots the logarithm of the CPI over the full sample. The series displays a clear and persistent upward trend with no tendency to revert to a fixed mean, which is the hallmark of a non-stationary process. Growth was steady and gradual from 2015 to early 2020, interrupted by a sharp but brief deflationary episode at the onset of the Covid-19 pandemic in spring 2020. From 2021 onward the series accelerated markedly, reflecting the post-pandemic inflation surge, before returning to a more moderate pace from 2023 onward.

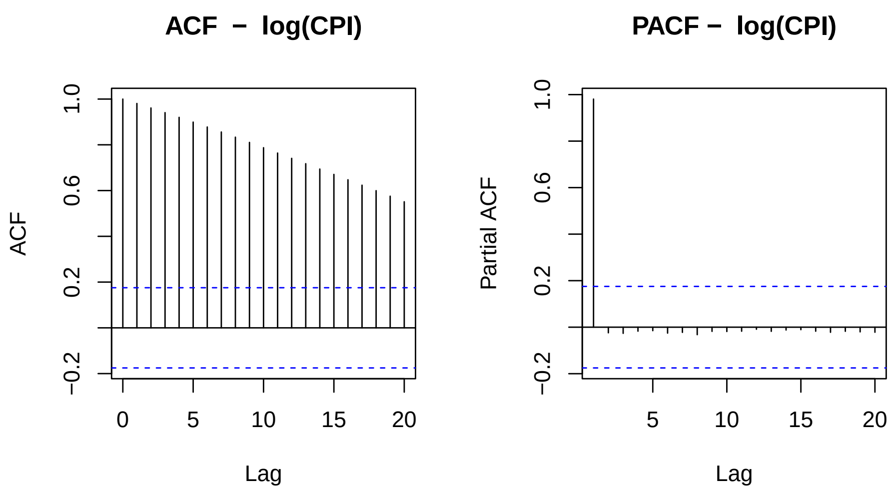

Figure 2: ACF and PACF of log CPI.

Figure 2 shows the ACF and PACF of the level. The ACF decays extremely slowly, remaining well above the confidence bands at lag 20, which is the textbook signature of a unit root. The PACF displays a single dominant spike at lag 1 close to unity, with all subsequent partial autocorrelations negligible. This pattern is characteristic of an I(1) process, or equivalently a random walk with drift.

Table 2: Descriptive statistics – log(CPI) and inflation

Series	T	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
log(CPI)	125	5.610	0.105	5.469	5.787	0.364	1.595
Inflation	124	0.254	0.272	−0.795	1.248	0.426	6.172

Inflation $\pi_t = 100\,\Delta\log(\mathrm{CPI}_t)$. Kurtosis is excess kurtosis; normal = 3.

Table 2 reports the descriptive statistics for both the level and the first difference. The standard deviation of log(CPI) is 0.105, reflecting the trend rather than fluctuations around a stable mean. The distribution is mildly right-skewed (0.364) and platykurtic (kurtosis 1.59), consistent with a smoothly trending series with no extreme observations.

Inflation

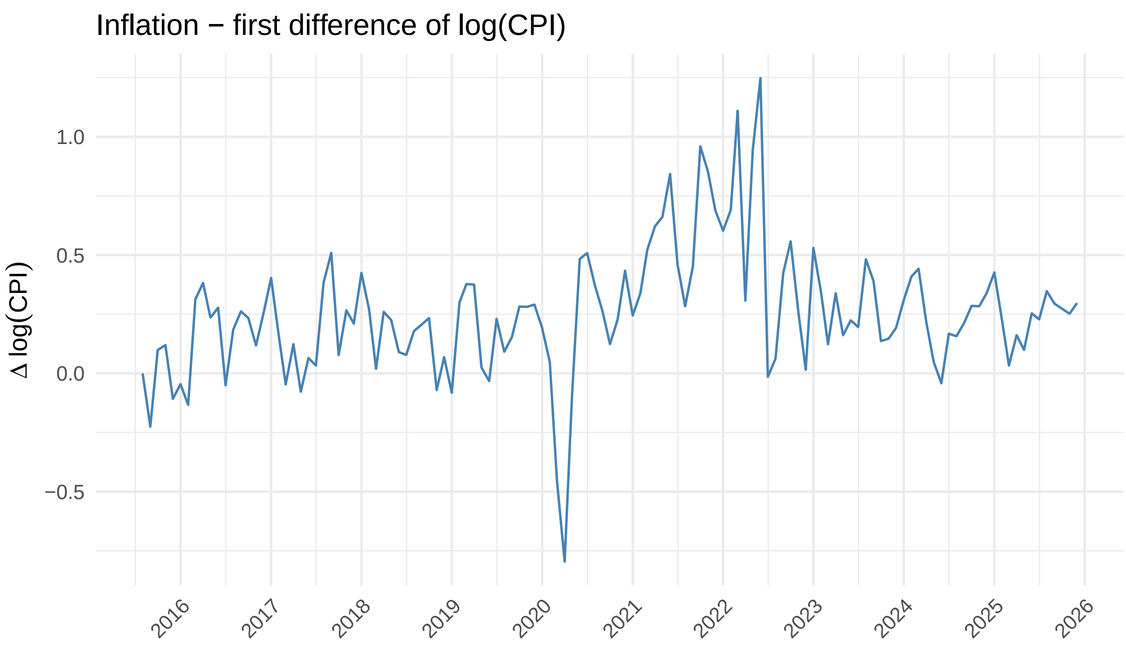

Figure 3: Inflation $\pi_t = 100\,\Delta\log(\mathrm{CPI}_t)$, August 2015 – December 2025.

Figure 3 plots the first difference of log(CPI), i.e. the monthly inflation rate. First differencing removes the trend entirely: the series now fluctuates around a positive mean of 0.25 percentage points per month with no visible drift, consistent with stationarity. The most striking features are the sharp deflationary spike in April 2020 (−0.80%) caused by the Covid-19 lockdowns, and the cluster of large positive values peaking in mid-2022 at the height of the post-pandemic inflation episode. The kurtosis of 6.17 confirms the presence of fat tails driven by these two episodes. The standard deviation of 0.27 is large relative to the mean, reflecting the exceptional volatility of the 2020–2022 period.

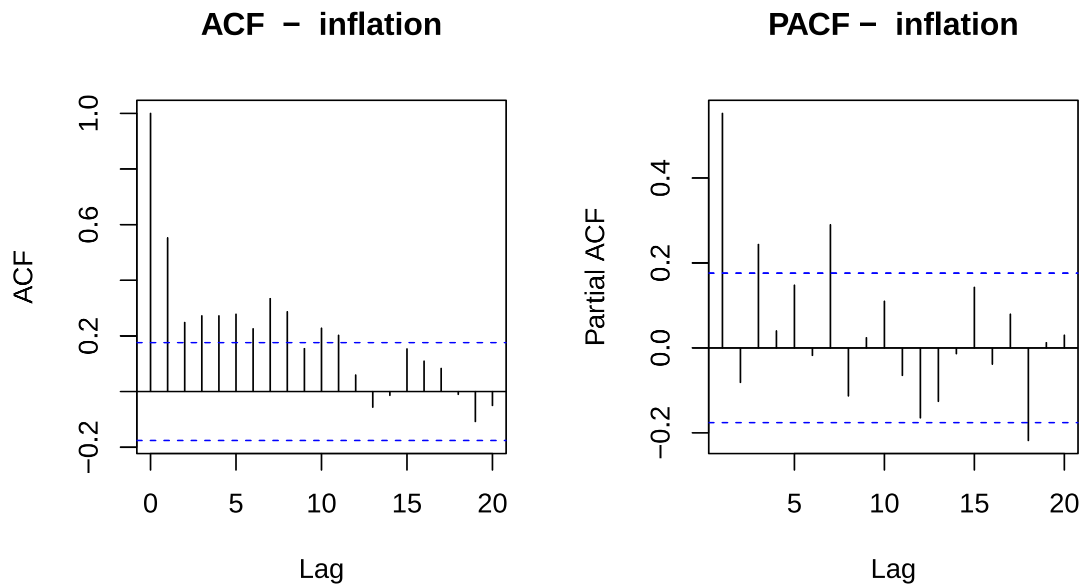

Figure 4: ACF and PACF of inflation.

Figure 4 shows the ACF and PACF of inflation. In sharp contrast to the level, the ACF drops immediately after lag 1 and remains mostly within the 95% confidence bands thereafter, confirming that first differencing is sufficient to achieve stationarity. The PACF shows significant spikes at lags 1 and 2, with sporadic significance at higher lags likely driven by the structural break induced by the 2020–2022 episode. This pattern suggests that a low-order AR process may provide a reasonable approximation for inflation, which we exploit in Question 2.

Log GDP

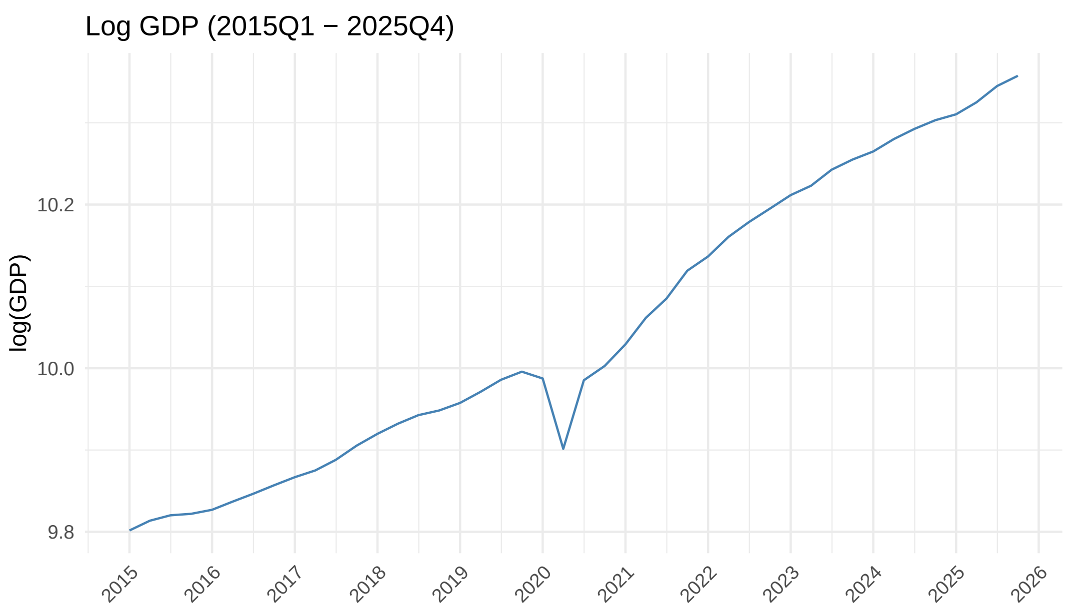

Figure 5: Log GDP, 2015Q1 – 2025Q4.

Figure 5 plots the logarithm of GDP over the sample. The series follows a smooth upward trend throughout, with one violent interruption: the Covid-19 recession of 2020Q2, which produced the sharpest quarterly contraction in modern US history, followed almost immediately by an equally sharp rebound in 2020Q3. Outside of this episode the series grows steadily and never reverts to a fixed mean, which is consistent with a non-stationary, trend-driven process.

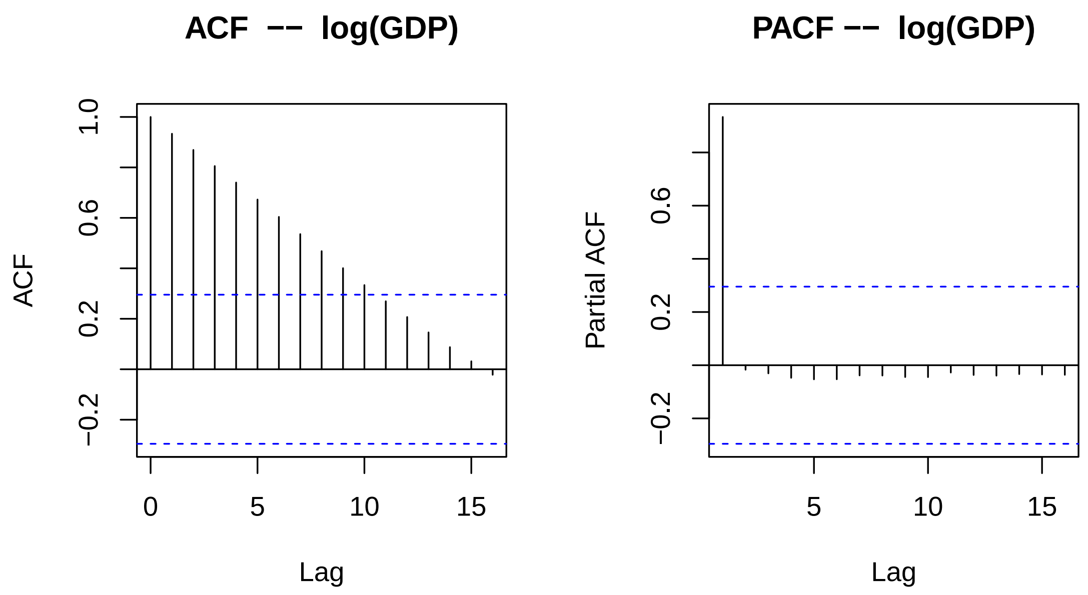

Figure 6: ACF and PACF of log GDP.

The ACF of the level (Figure 6) decays very slowly and remains high across all reported lags, replicating the same unit root signature observed for log CPI. The PACF concentrates almost all explanatory power in a single spike at lag 1, with negligible partial autocorrelations thereafter. The series is I(1).

Table 3: Descriptive statistics – log(GDP) and GDP growth

Series	T	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
log(GDP)	44	10.046	0.178	9.802	10.357	0.299	1.670
GDP growth	43	5.170	8.080	−34.4	33.5	−1.760	17.800

GDP growth = $400\,\Delta\log(\mathrm{GDP}_t)$, annualised percentage points. Kurtosis is excess kurtosis; normal = 3.

Table 3 reports the descriptive statistics. The standard deviation of log(GDP) is 0.178, again reflecting the trend rather than genuine volatility around a mean. The distribution is mildly right-skewed (0.299) and platykurtic (kurtosis 1.67), consistent with a smoothly trending series.

GDP growth rate

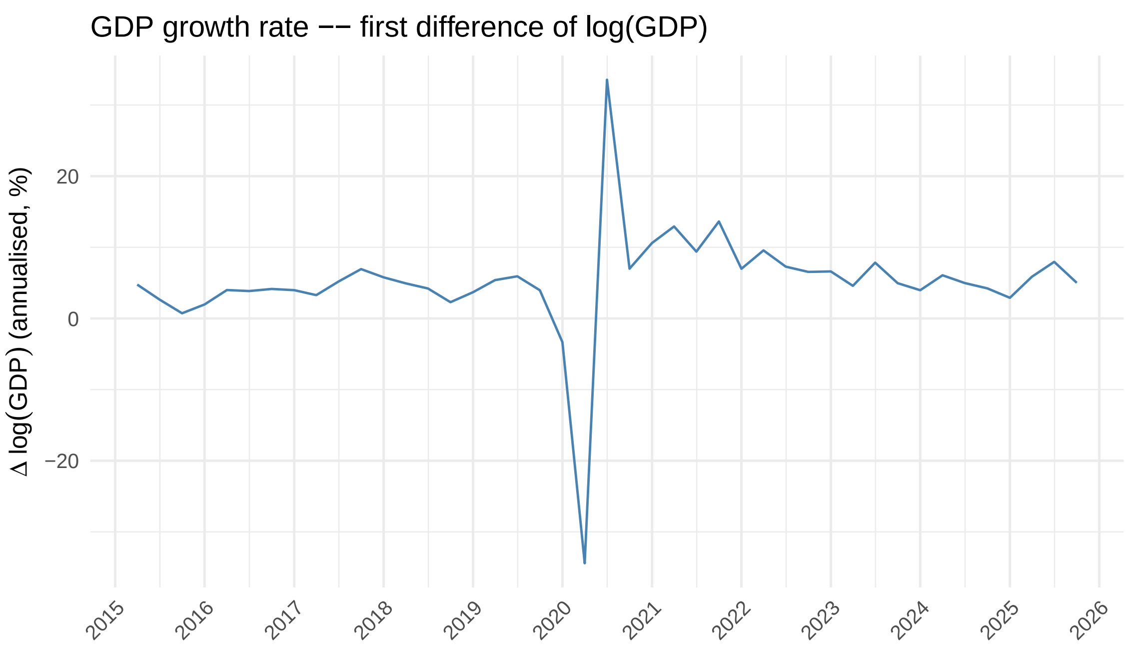

Figure 7: GDP growth rate $400\,\Delta\log(\mathrm{GDP}_t)$, annualised percentage points, 2015Q2 – 2025Q4.

Taking the first difference and scaling by 400 yields the annualised quarterly growth rate (Figure 7). The series is now stationary, fluctuating around a mean of 5.2% with no visible trend. The Covid episode dominates the picture entirely: the 2020Q2 observation of −34.4% and the subsequent rebound of +33.5% are extreme outliers by any standard. These two observations are responsible for the highly negative skewness (−1.76) and the exceptional kurtosis of 17.8, which is roughly six times the Gaussian benchmark. Excluding the Covid quarter, the series behaves much more regularly, with moderate fluctuations consistent with normal business cycle dynamics.

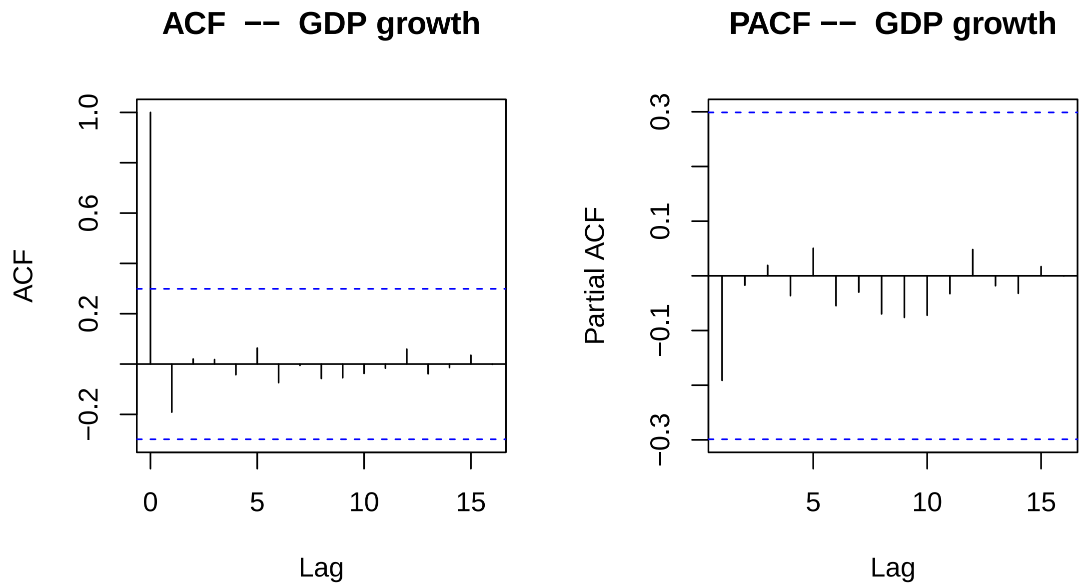

Figure 8: ACF and PACF of GDP growth.

The ACF of GDP growth (Figure 8) drops sharply after lag 1 and remains within the 95% confidence bands at all subsequent lags, confirming stationarity. The PACF shows no clearly significant spike beyond lag 1, suggesting that a low-order AR process may be sufficient to capture the serial dependence in this series, which we investigate in Question 2.

Federal Funds Rate

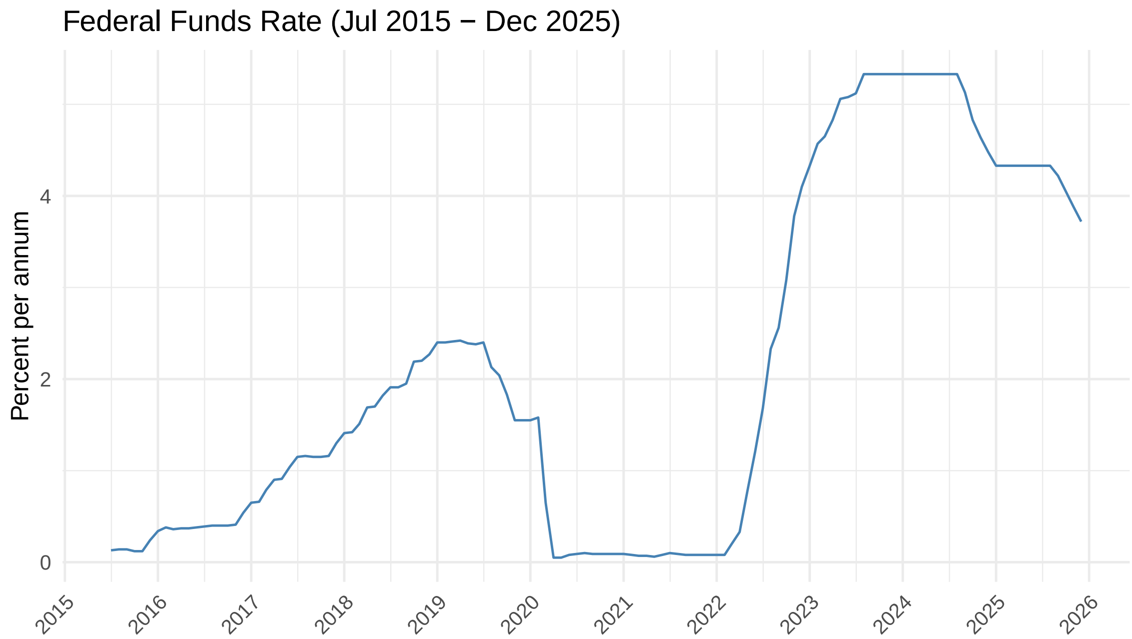

Figure 9: Federal Funds Rate, July 2015 – December 2025.

Figure 9 plots the Federal Funds effective rate over the sample. Unlike the price level and GDP, this series does not display a monotone trend but rather moves in discrete policy-driven cycles. The rate was gradually declining toward the zero lower bound in 2015–2016, remained near zero through the Covid shock of 2020, then rose sharply from early 2022 onward as the Federal Reserve embarked on the most aggressive tightening cycle in decades, reaching a peak of 5.33% in mid-2023. A gradual easing cycle began in late 2024. The series mean over the sample is 2.08% with a standard deviation of 1.92%, reflecting the wide range of policy stances covered by the sample.

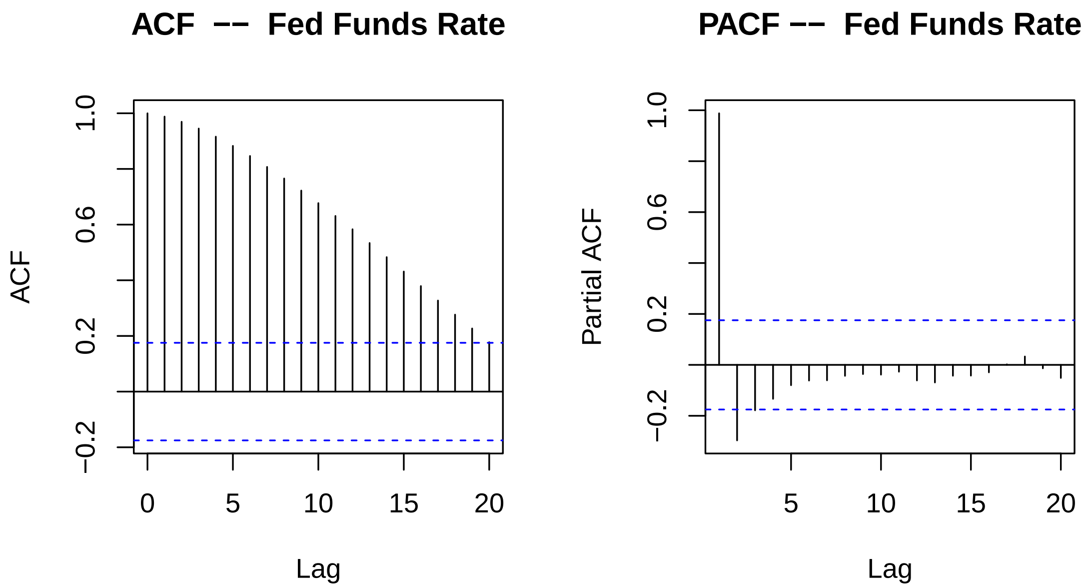

Figure 10: ACF and PACF of the Federal Funds Rate.

The ACF of the level (Figure 10) decays slowly and remains significantly positive at long lags, while the PACF is dominated by a single spike at lag 1. This pattern is consistent with a near-unit-root or highly persistent process. Whether the rate is truly I(1) or stationary around a slowly shifting mean is a well-known empirical debate; for the purposes of this exercise we treat it as I(1) and proceed to first-difference.

Table 4: Descriptive statistics – Federal Funds Rate and its first difference

Series	T	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
Fed Funds Rate	125	2.080	1.920	0.050	5.330	0.552	1.780
Δ Fed Funds Rate	124	0.029	0.185	−0.930	0.700	−0.460	11.100

Rate in percent per annum, not seasonally adjusted. Kurtosis is excess kurtosis; normal = 3.

First difference of the Federal Funds Rate

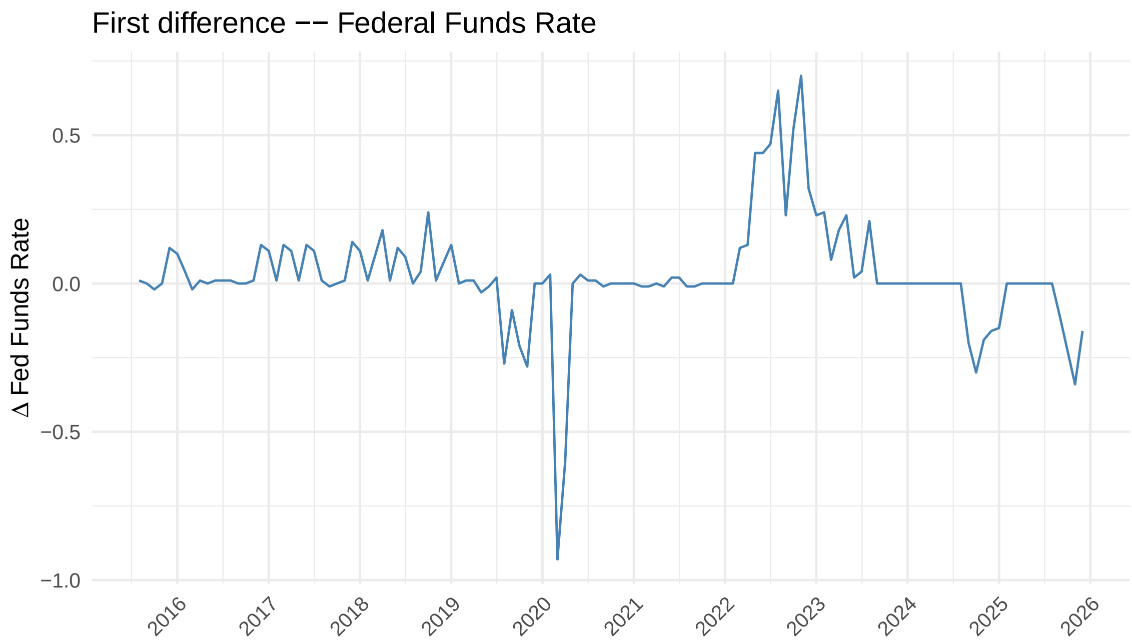

Figure 11: First difference of the Federal Funds Rate, August 2015 – December 2025.

The first difference (Figure 11) fluctuates around zero and is clearly more stationary than the level. The series is dominated by large discrete jumps that correspond directly to FOMC decisions: the emergency cut of March 2020 (−0.93 percentage points) and the rapid sequence of hikes in 2022–2023 stand out prominently. The kurtosis of 11.1 is a direct consequence of these large but infrequent moves, which generate fat tails relative to a Gaussian distribution. The mild negative skewness (−0.46) reflects the asymmetry between the sharp emergency cut and the more gradual tightening.

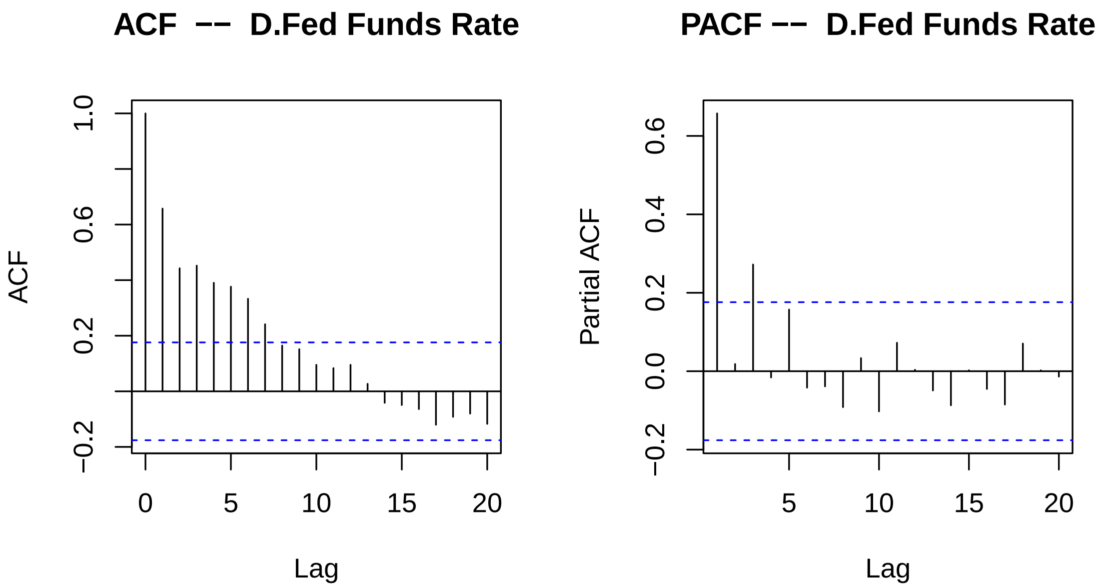

Figure 12: ACF and PACF of the first difference of the Federal Funds Rate.

The ACF of the first difference (Figure 12) declines more quickly than in the level, though some persistence remains visible. The PACF displays a significant spike at lag 1 and possibly lag 2, suggesting that changes in the policy rate are somewhat predictable from their own recent history — a reflection of the gradual and pre-announced nature of Fed tightening and easing cycles. A low-order AR model for the differenced rate should capture this dependence adequately.

Daily electricity demand

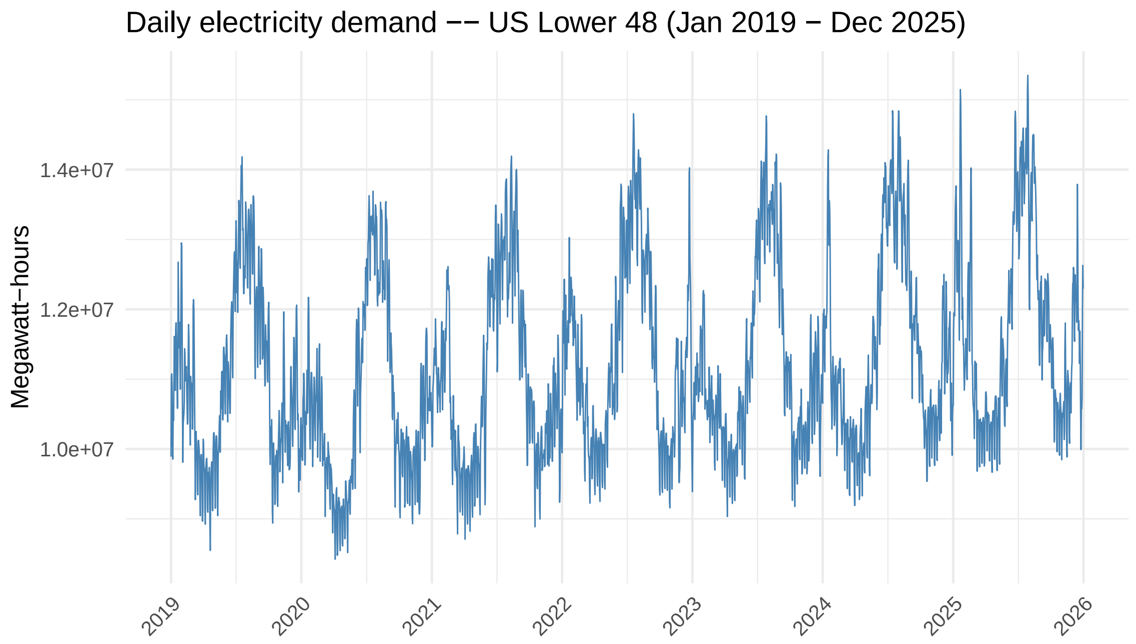

Figure 13: Daily electricity demand, US Lower 48, January 2019 – December 2025.

Figure 13 plots daily electricity demand for the contiguous United States over the full sample. The series differs markedly from the macroeconomic variables studied above. There is no clear deterministic trend over the 2019–2025 window; demand fluctuates between approximately 8.4 and 15.3 million MWh throughout the sample with no tendency to drift upward or downward. However, the series is far from stationary: it exhibits strong and regular seasonal oscillations that repeat every year, with peaks in summer (driven by air-conditioning load) and winter (heating demand), and troughs in the mild shoulder seasons of spring and autumn. This is seasonal non-stationarity, which is qualitatively different from the stochastic trend non-stationarity observed in log CPI and log GDP. The Covid-19 period is also visible as a modest dip in spring 2020, reflecting the reduction in commercial and industrial activity during lockdowns, though the effect is far smaller than in the GDP series. The sample mean is 11.2 million MWh with a standard deviation of 1.35 million MWh; the positive skewness of 0.576 reflects the tendency for summer and winter peaks to be more extreme than the seasonal troughs.

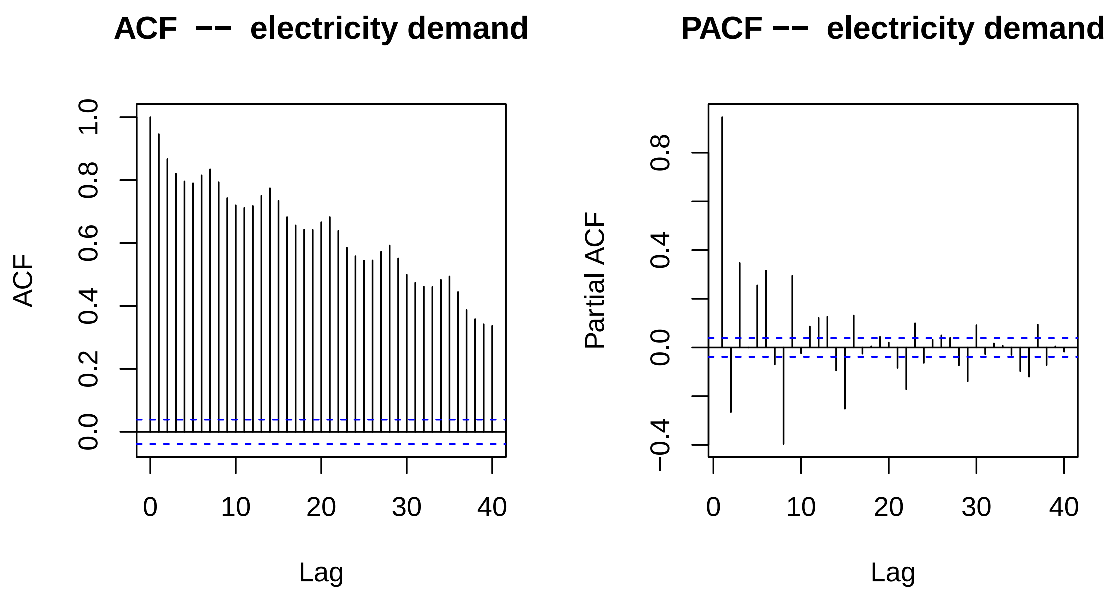

Figure 14: ACF and PACF of daily electricity demand (40 lags).

The ACF of the level (Figure 14) reveals two distinct layers of dependence. First, it decays slowly from unity, reflecting the strong seasonal persistence of the series — adjacent days within the same season have very similar demand levels. Second, and more strikingly, the ACF oscillates in a wave pattern and remains significantly positive at lags 7, 14, 21, and 28. This is the textbook signature of **weekly seasonality**: electricity demand on any given day is more similar to demand on the same day of the previous week than to demand on the preceding day, because weekdays and weekends have systematically different consumption profiles. The PACF concentrates this dependence in large spikes at lags 1 and 7, with a secondary spike at lag 2, confirming that the dominant short-run dynamics are a combination of day-to-day persistence and a 7-day seasonal cycle.

Table 5: Descriptive statistics – daily electricity demand and its first difference

Series	T	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
Electricity demand	2557	11,200,000	1,350,000	8,420,000	15,300,000	0.576	2.550
Δ Elec. demand	2556	940	443,000	$-1{,}670{,}000$	1,890,000	0.349	3.490

All values in megawatt-hours (MWh). Kurtosis is excess kurtosis; normal = 3.

First difference of electricity demand

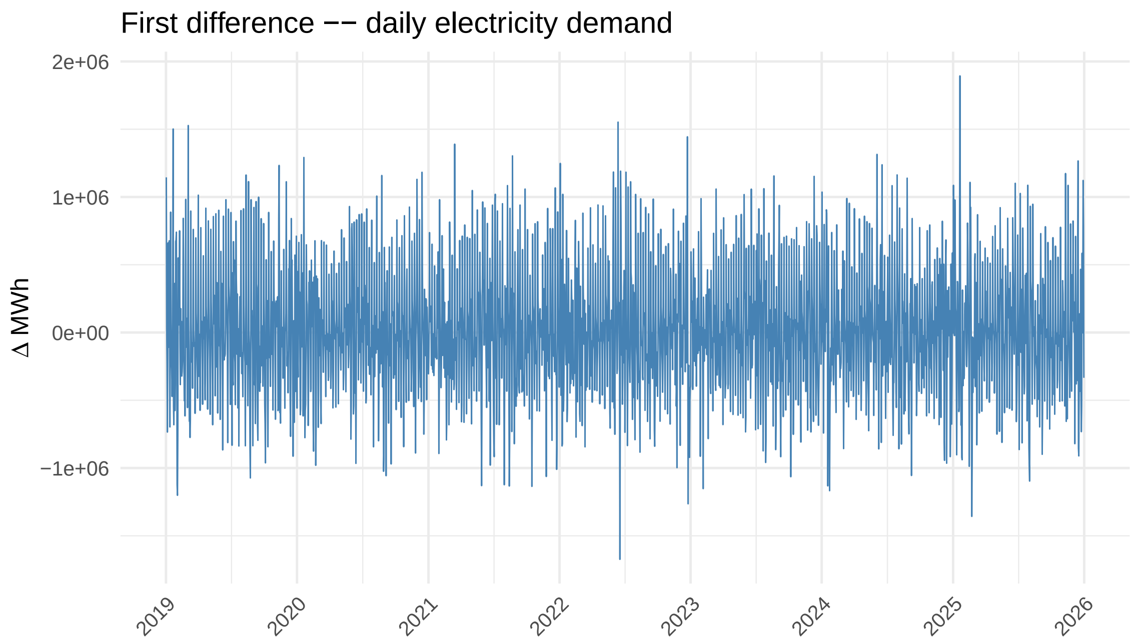

Figure 15: First difference of daily electricity demand, January 2019 – December 2025.

Simple first differencing removes the slow seasonal drift but does not eliminate the weekly pattern (Figure 15). The differenced series fluctuates around zero with a negligible mean of 940 MWh — essentially zero relative to the level — but its variance is enormous, with daily changes ranging from −1.67 to +1.89 million MWh. The large swings visible throughout the sample are the mechanical consequence of the weekly cycle: the transition from Sunday to Monday (or Friday to Saturday) produces a large positive (or negative) change in demand every single week. The kurtosis of 3.49 is close to the Gaussian benchmark of 3, suggesting that once the weekly structure is accounted for, the residual distribution is approximately normal.

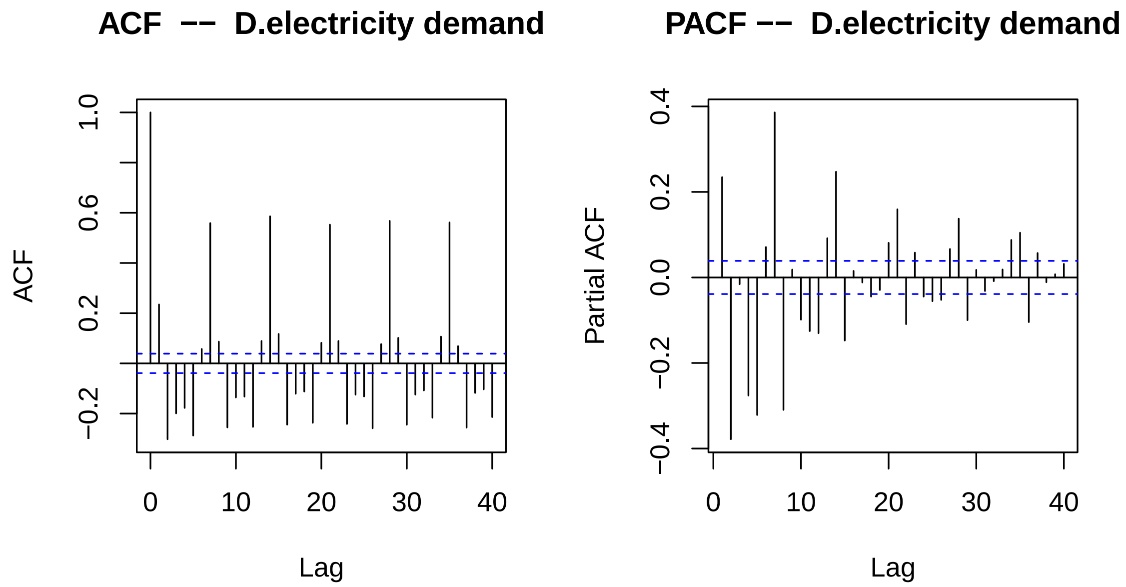

Figure 16: ACF and PACF of the first difference of electricity demand (40 lags).

The ACF of the first difference (Figure 16) confirms that the weekly seasonality survives simple differencing: significant spikes remain at lags 7, 14, 21, and 28, with the ACF oscillating between positive and negative values in a pattern driven entirely by the day-of-week cycle. The PACF similarly retains spikes at lags 1 and 7.

The PACF similarly retains spikes at lags 1 and 7. This suggests that ordinary first differencing is not sufficient to remove the seasonal structure. As a robustness check, I therefore also consider the seasonal difference $y_t - y_{t-7}$, which directly targets the weekly cycle in daily demand and is more appropriate for a series with strong day-of-week seasonality.

1.3 Best AR(p) process for each variable

The goal is to find, for each series, the AR(p) model of the form

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t, \qquad \varepsilon_t \sim \mathrm{WN}(0, \sigma^2),$$

that best describes the serial dependence in the data. All models are estimated by OLS.

A necessary condition for an AR(p) to be well-specified is that the process be covariance-stationary. The analysis in Question 1 showed that log(CPI), log(GDP), and the Federal Funds Rate are all I(1): their ACFs decay slowly and their PACFs are dominated by a spike at lag 1 close to unity. We therefore apply AR(p) models to their first differences: the inflation rate $\pi_t = 100\,\Delta\log(\mathrm{CPI}_t)$, the annualised GDP growth rate $400\,\Delta\log(\mathrm{GDP}_t)$, and Δr_t, rather than to the levels. Inflation is already stationary and is modelled directly. For daily electricity demand, simple first differencing removes the slow seasonal drift, and we include lags up to order 14 to ensure the weekly seasonal cycle at lag 7 is adequately captured.

The lag order p is selected by minimizing the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) over a grid of candidate models. Because BIC penalizes model complexity more heavily than AIC, the two criteria sometimes select different orders; when they disagree we report both and favor the BIC solution as the benchmark, given its consistency property. The maximum lag orders considered are 12 for monthly series, 8 for the quarterly GDP series, and 14 for the daily electricity series.

Table 6: Lag order selection summary

Series	Frequency	Estimated on	$p_{\max}$	AIC	BIC	Selected
Inflation	Monthly	π_t	12	8	3	AR(3)
GDP growth	Quarterly	$400\,\Delta\log(\mathrm{GDP}_t)$	8	8	8	AR(8)
Fed Funds Rate	Monthly	Δr_t	12	3	1	AR(1)
Electricity demand	Daily	$\Delta \mathrm{Elec}_t$	14	14	14	AR(14)

When AIC and BIC disagree, the BIC selection is adopted as benchmark.

Lag order selection. Figures 17–20 plot the AIC and BIC as functions of p for each series.

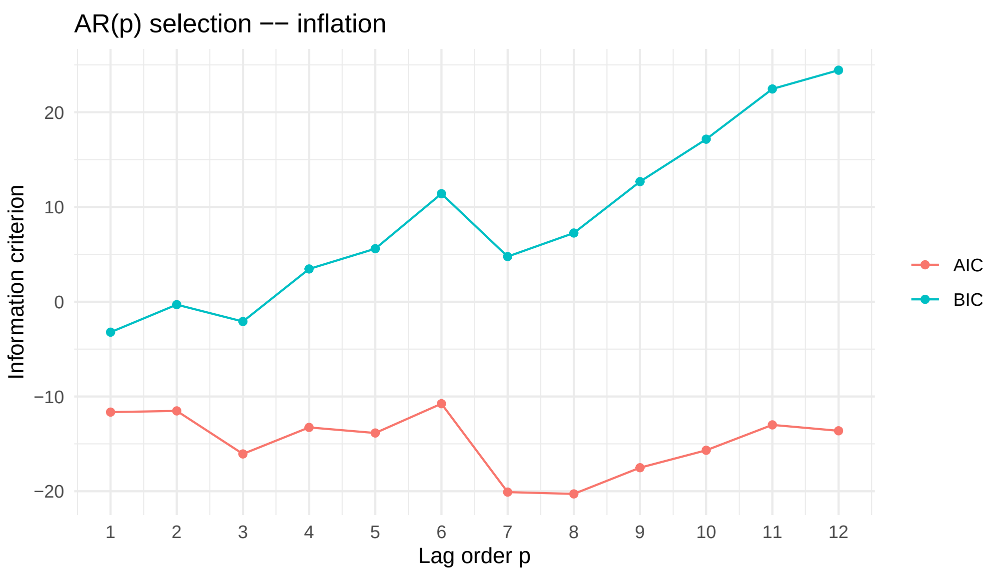

Figure 17: AR(p) lag selection – inflation.

For inflation, BIC reaches its minimum at $p = 3$ while AIC continues declining to $p = 8$. The two criteria disagree, and we select AR(3) on the basis of BIC.

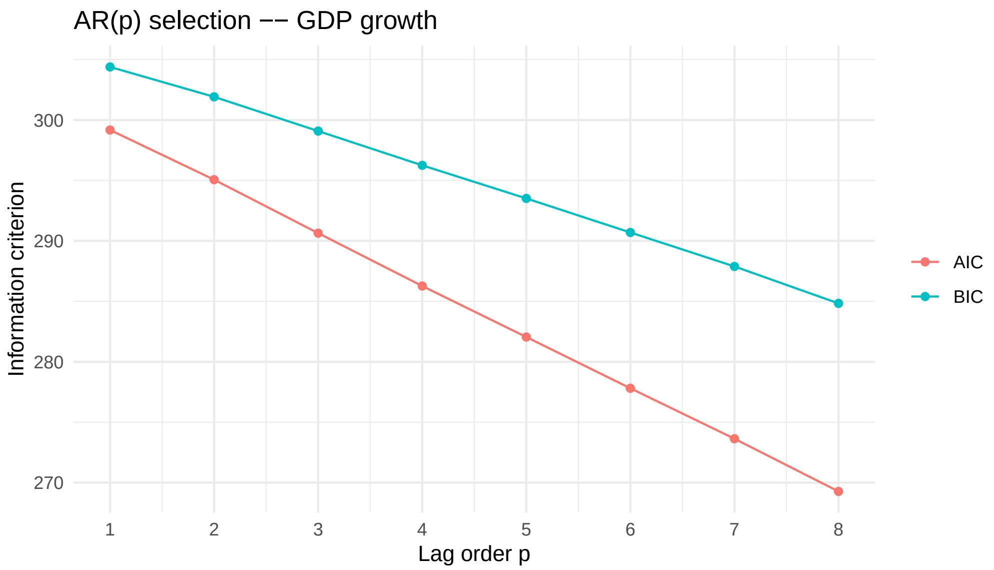

Figure 18: AR(p) lag selection – GDP growth.

For GDP growth, both AIC and BIC decrease monotonically across the entire grid up to $p = 8$, which is the maximum order considered given the short quarterly sample of 43 observations. Both criteria therefore select AR(8), though we note that the short sample limits the reliability of higher-order estimates.

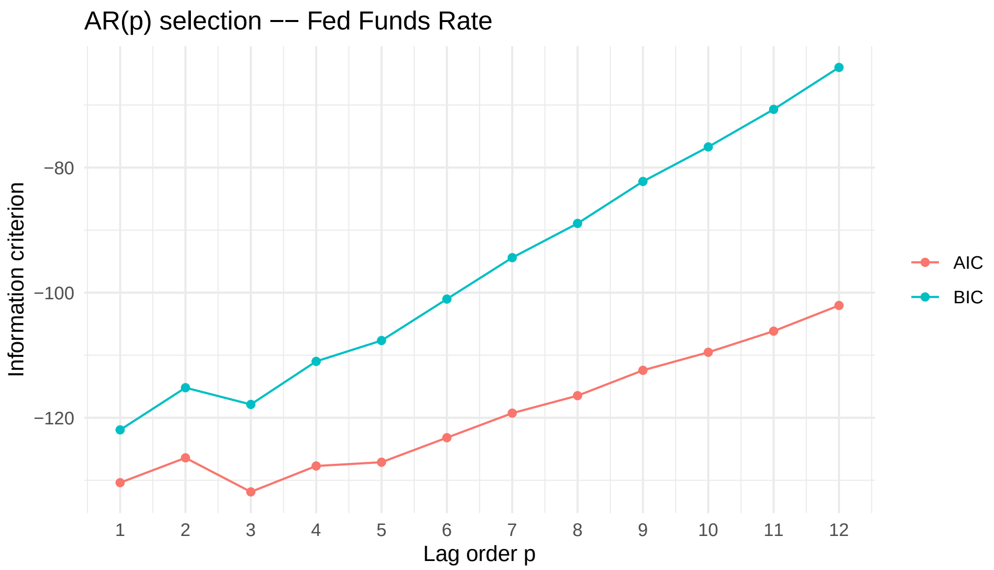

Figure 19: AR(p) lag selection – Δ Federal Funds Rate.

For the first difference of the Federal Funds Rate, AIC selects $p = 3$ while BIC selects $p = 1$. Following our stated rule we adopt AR(1) as the benchmark model.

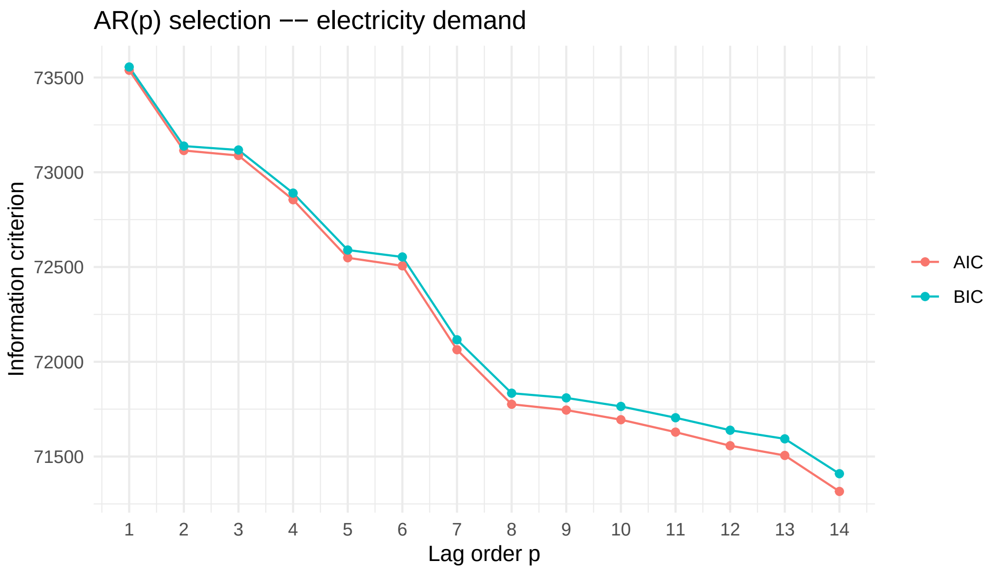

Figure 20: AR(p) lag selection – Δ electricity demand.

For electricity demand, both AIC and BIC decrease monotonically and reach their minimum at the boundary $p = 14$. The selection is consistent with the strong weekly seasonality identified in Question 1: lags at multiples of 7 are needed to capture the day-of-week cycle, and $p = 14$ covers exactly two full weekly cycles.

Since both criteria reach their minimum at the boundary of the search grid, the selected order should be interpreted as a lower bound; the residual diagnostics nonetheless confirm that 14 lags are sufficient to whiten the residuals at conventional significance levels.

Estimation results. Table 7 reports the OLS estimates for all four selected models.

Table 7: OLS estimates of selected AR(p) models

	Inflation AR(3)	GDP growth AR(8)	Δ Fed Funds AR(1)	Δ Elec. demand AR(14)
Constant	0.098^{**} (0.031)	$8.015^{\cdot}$ (3.937)	0.009 (0.013)	−52.6 (5912.9)
ϕ_1	0.608^{***} (0.090)	−0.216 (0.195)	0.663^{***} (0.069)	0.310^{***} (0.019)
ϕ_2	-0.219^{*} (0.103)	−0.039 (0.200)		-0.311^{***} (0.020)
ϕ_3	0.242^{**} (0.088)	0.000 (0.199)		-0.047^{*} (0.021)
ϕ_4		−0.041 (0.199)		-0.124^{***} (0.021)
ϕ_5		0.032 (0.198)		-0.118^{***} (0.021)
ϕ_6		−0.065 (0.197)		-0.128^{***} (0.021)
ϕ_7		−0.045 (0.197)		0.317^{***} (0.021)
ϕ_8		−0.071 (0.194)		-0.302^{***} (0.021)
ϕ_9				0.042^{*} (0.021)
ϕ_{10}				-0.059^{**} (0.021)
ϕ_{11}				−0.033 (0.021)
ϕ_{12}				-0.077^{***} (0.021)
ϕ_{13}				0.011 (0.020)
ϕ_{14}				0.250^{***} (0.019)
T	121	35	123	2,542
R^2	0.347	0.058	0.436	0.547
Adj. R^2	0.330	−0.232	0.431	0.545
$\hat{\sigma}$	0.221	9.882	0.140	298,100
Selected by	BIC	AIC & BIC	BIC	AIC & BIC

Standard errors in parentheses. Significance: $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Inflation and Fed Funds Rate are estimated on first differences. GDP growth $= 400\,\Delta\log(\mathrm{GDP}_t)$.

Inflation – AR(3). All three lags are statistically significant. The first lag carries the largest coefficient ($\hat{\phi}_1 = 0.608$), reflecting strong month-to-month momentum in inflation. The second lag enters negatively ($\hat{\phi}_2 = -0.219$), introducing a mild corrective dynamic, while the third lag is again positive ($\hat{\phi}_3 = 0.242$). The model explains 33% of the variance in inflation (adjusted R^2), which is reasonable for a monthly macroeconomic series. The F-test strongly rejects the null that all coefficients are zero ($p < 10^{-10}$).

GDP growth – AR(8). Although the information criteria select an AR(8), this specification should be interpreted with caution. In the estimated model, none of the eight lag coefficients is individually significant, the overall F-test has a p-value of 0.988, and the adjusted R^2 is negative (-0.23). This does not necessarily imply that GDP growth is unpredictable; rather, it suggests that an AR(8) is too parameter-rich for the available sample. After differencing and including eight lags, only 35 effective observations remain, so estimating nine parameters leaves just 26 degrees of freedom. In addition, the extreme Covid contraction in 2020Q2 (-34.4%) likely inflates the residual variance and standard errors. For these reasons, the AR(8) selected mechanically by the criteria is probably overfitted, and a more parsimonious specification—such as an AR(1) or even a constant-only model—is more credible for quarterly GDP growth in this sample.

Δ *Federal Funds Rate – AR(1).* The single lag is highly significant ($\hat{\phi}_1 = 0.663$, $t = 9.67$), capturing the strong persistence of FOMC decisions: a rate change in one month predicts the direction of the next. The model explains 43% of the variance. The intercept is not significant, consistent with the changes being mean-zero on average over the full cycle.

Δ *Electricity demand – AR(14).* The electricity model is the most interesting. Lags 1, 2, 7, 8, and 14 are all highly significant ($p < 0.001$), and the pattern is striking: $\phi_1 > 0$ and $\phi_2 < 0$ capture short-run mean reversion within the week, while $\phi_7 > 0$ and $\phi_8 < 0$ replicate the same pattern one week later, and $\phi_{14} > 0$ extends it to a two-week horizon. This is the AR representation of the weekly seasonal cycle identified in the ACF of Question 1. Lags 4–6 are also significant and negative, reflecting the within-week decline from Monday to Friday. The model explains 54.5% of the variance in daily demand changes with 2,527 degrees of freedom, and the F-test is overwhelming ($p < 2.2 \times 10^{-16}$).

As a robustness check, we also estimated an AR model for the seasonally differenced electricity series, $y_t - y_{t-7}$, which directly targets the weekly cycle visible in the daily data. The information criteria again selected an AR(14), but this specification performs substantially better than the simple first-difference model: the adjusted R^2 rises from 0.545 to 0.847, while the AIC falls from 71316 to 71239 and the BIC from 71409 to 71332. This confirms that weekly seasonality is an important feature of electricity demand and that seasonal differencing provides a more appropriate robustness specification.

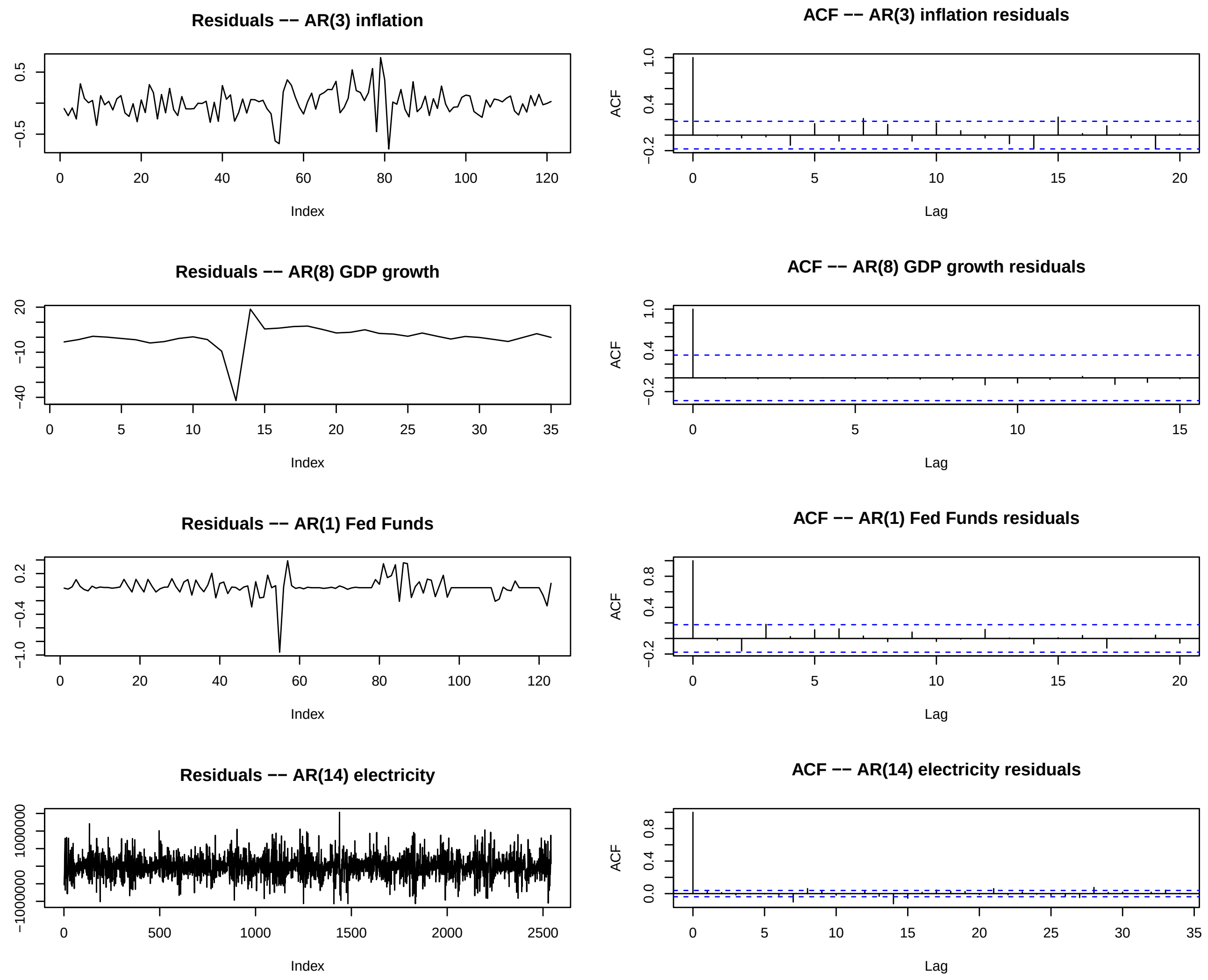

Figure 21: Residuals and ACF of residuals for the four selected AR models.

Residual diagnostics. Figure 21 shows the residual series and their ACFs for all four models. For inflation and the Fed Funds Rate the residual ACFs are clean, with no spike outside the 95% confidence bands — the selected orders are adequate. For GDP growth the residual ACF also shows no significant autocorrelation, though the single large outlier (Covid quarter) is clearly visible in the residual plot and dominates the scale. For electricity demand the residual ACF of the first-difference AR(14) model is mostly clean, though a marginal spike at lag 7 remains visible, suggesting that the weekly cycle is not entirely absorbed by 14 lags. A robustness check using the seasonal difference $y_t - y_{t-7}$ produces a noticeably better fit and cleaner diagnostics, confirming that explicit treatment of weekly seasonality improves the electricity specification.

1.4 1-step and 3-step ahead forecasts

Having selected and estimated the best AR(p) model for each series, we now produce out-of-sample forecasts at horizons $h = 1$ and $h = 3$. The forecast origin is the last available observation in each dataset: December 2025 for the monthly series, 2025Q4 for GDP, and 31 December 2025 for electricity demand.

Methodology. For a stationary AR(p) process, the optimal h-step ahead forecast under quadratic loss is obtained by iterating the model forward. The 1-step ahead forecast is

$$\hat{y}_{T+1|T} = \hat{c} + \hat{\phi}_1 y_T + \cdots + \hat{\phi}_p y_{T-p+1},$$

and the h-step ahead forecast substitutes previously computed forecasts for unobserved future values:

$$\hat{y}_{T+h|T} = \hat{c} + \hat{\phi}_1 \hat{y}_{T+h-1|T} + \cdots + \hat{\phi}_p \hat{y}_{T+h-p|T},$$

where $\hat{y}_{T+j|T} = y_{T+j}$ for $j \leq 0$ since those values are observed. This iterated forecasting procedure is implemented via a custom `ar_forecast()` function in R.

For the three series modelled in first differences — GDP growth, the Federal Funds Rate, and electricity demand — the AR forecasts are for Δy_{T+h}. We recover level forecasts by cumulating the predicted changes from the last observed level:

$$\hat{y}_{T+h|T} = y_T + \sum_{j=1}^{h} \widehat{\Delta y}_{T+j|T}.$$

Table 8 collects all forecasts and Figures 22–25 plot them against the recent history of each series.

Table 8: 1-step and 3-step ahead forecasts

Series	Last observed	$h = 1$	$h = 3$
Inflation (pp/month)	0.297	0.295	0.272
GDP growth (ann. %)	5.010	5.761	5.734
log(GDP)	10.357	10.372	10.400
Fed Funds Rate (%)	3.720	3.623	3.539
Electricity demand (MWh)	12,293,764	11,754,287	11,639,370

Forecasts produced by iterated OLS-AR models. Last observed values are for December 2025 (monthly), 2025Q4 (quarterly), and 31 December 2025 (daily). $h = 1$ and $h = 3$ denote one- and three-step ahead horizons respectively.

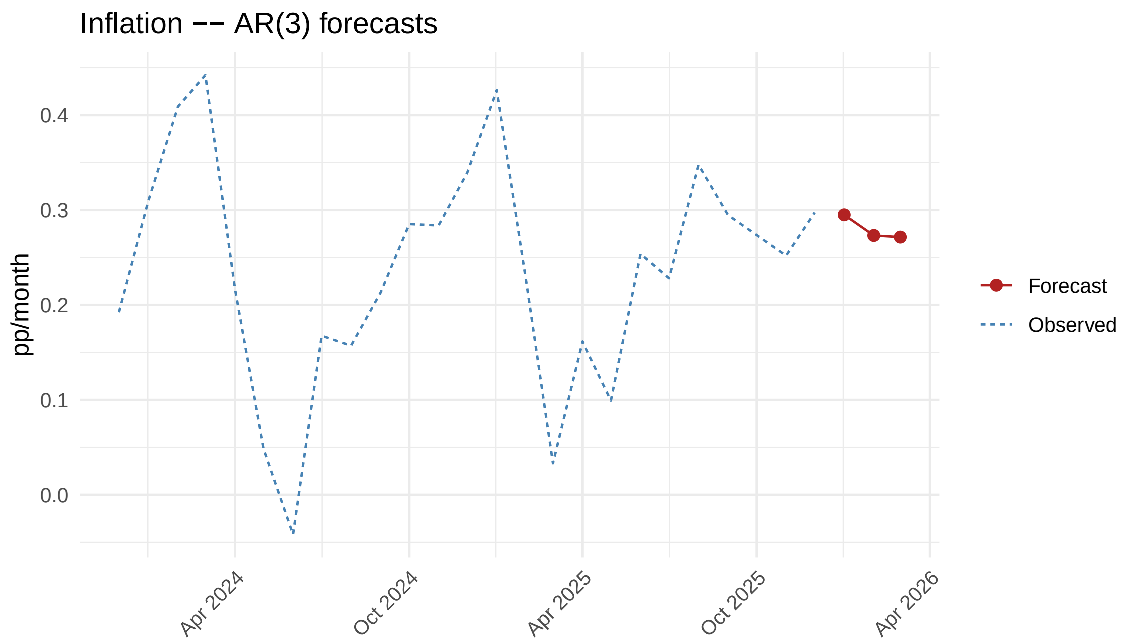

Figure 22: Inflation: AR(3) forecasts at $h = 1$ and $h = 3$. The last 24 observed months are shown in blue; forecasts in red.

Inflation – AR(3). The AR(3) model predicts inflation of 0.295 percentage points per month at the 1-step horizon, barely below the last observed value of 0.297. The 3-step forecast eases slightly to 0.272 pp/month, corresponding to roughly 3.3% annualised. This mild downward path is consistent with the gradual disinflation visible in the data since mid-2023: the model captures the recent trend of inflation returning toward its long-run mean, but does so slowly because $\hat{\phi}_1 = 0.608$ implies strong persistence. The forecasts are tightly clustered around the recent observed values, reflecting the fact that monthly inflation has been stable and low-volatility since 2024.

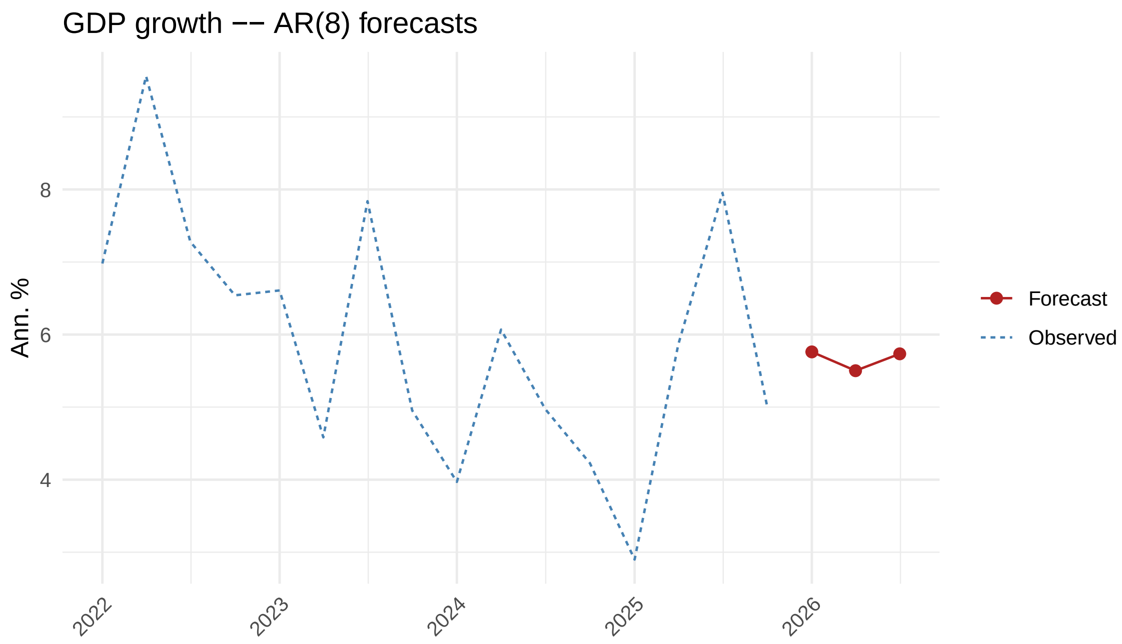

Figure 23: GDP growth: AR(8) forecasts at $h = 1$ and $h = 3$. The last 16 observed quarters are shown in blue; forecasts in red.

GDP growth – AR(8). The AR(8) model predicts annualized GDP growth of 5.76% at the 1-step horizon and 5.73% at the 3-step horizon. As discussed in Question 2, none of the eight lags were individually significant, so the forecasts are driven almost entirely by the estimated intercept of 8.0, which captures the unconditional mean growth rate inflated by the Covid rebound. The near-identical 1- and 3-step forecasts reflect this: with no meaningful autoregressive dynamics, the model simply reverts immediately to its mean. The log(GDP) forecasts of 10.372 and 10.400 imply a continued expansion of the US economy at a moderate pace, consistent with the 2024–2025 trend visible in the plot. These forecasts should be interpreted with caution given the overparameterisation noted in Question 2.

As a robustness check, an AR(1) model for GDP growth yields nearly identical forecasts (5.14% at $h = 1$, 5.12% at $h = 3$), confirming that the point forecasts are not materially sensitive to the choice of lag order.

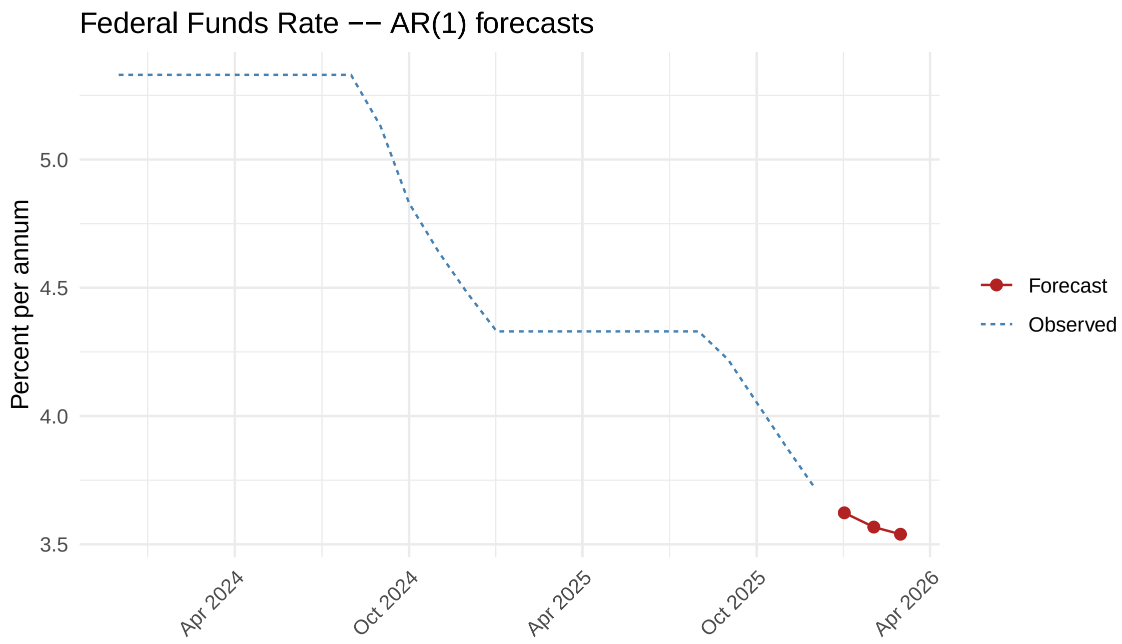

Figure 24: Federal Funds Rate: AR(1) forecasts at $h = 1$ and $h = 3$. The last 24 observed months are shown in blue; forecasts in red.

Federal Funds Rate – AR(1). The AR(1) model on the first-differenced rate predicts continued easing: the rate is forecast to fall from its last observed value of 3.72% to 3.62% at the 1-step horizon and 3.54% at the 3-step horizon. The dynamics are driven by $\hat{\phi}_1 = 0.663$, which implies that the direction of the most recent change, a cut of roughly 0.10 pp in December 2025, is expected to persist but attenuate over the following months. This is economically coherent: the Federal Reserve was in an easing cycle at the end of the sample, and the model correctly extrapolates this gradual descent. The forecasts suggest the rate will be approaching 3.5% by March 2026, which is consistent with market expectations at the time of writing.

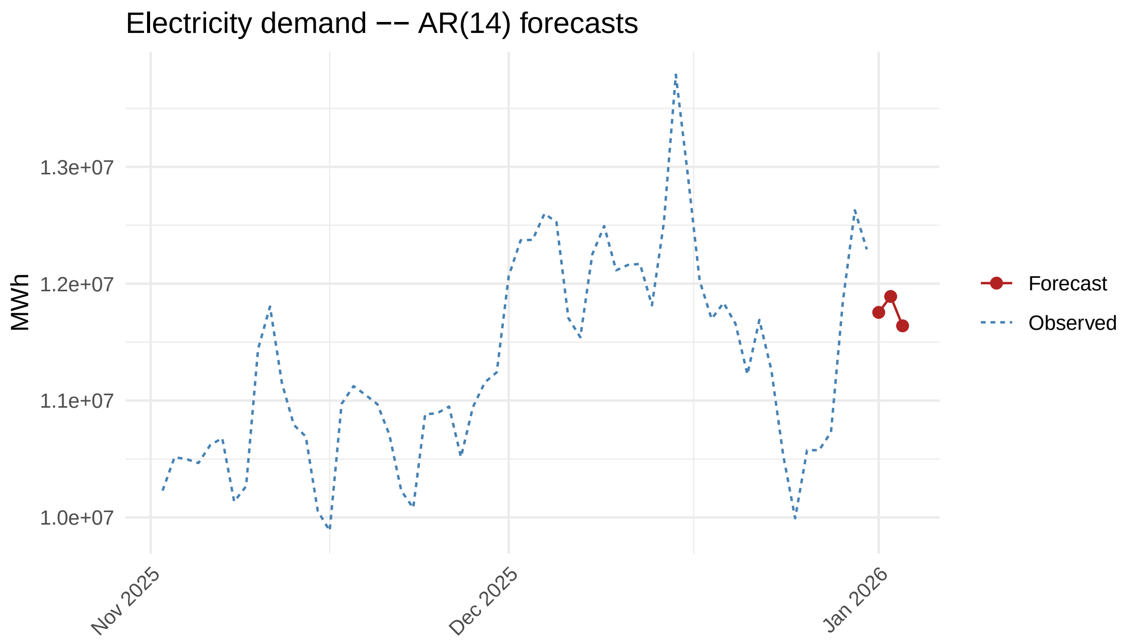

Figure 25: Electricity demand: AR(14) forecasts at $h = 1$ and $h = 3$. The last 60 observed days are shown in blue; forecasts in red.

Electricity demand – AR(14). The electricity forecasts are the most distinctive. The 1-step ahead forecast of 11.75 million MWh represents a sharp drop of roughly 540,000 MWh from the last observed value of 12.29 million MWh on 31 December 2025. This is not a model failure: 31 December was a Wednesday with high demand, while 1 January is a public holiday with systematically lower consumption. The AR(14) model captures this through the large negative coefficients at lags 4–6 and the positive coefficient at lag 7, which together encode the weekly seasonal pattern identified in Question 1. The 3-step forecast of 11.64 million MWh for 3 January 2026 (a Saturday) is further reduced, again consistent with lower weekend demand. The visible oscillation in the forecast reflects the weekly cycle being projected forward, exactly what a well-specified AR model with sufficient lags should do.

A seasonal-difference robustness specification produces very similar 1-step and 3-step ahead forecasts, indicating that the short-run forecast conclusions are not materially affected by the choice between simple first differencing and seasonal differencing.

2 Problem II

Data

For Problem II we use United States macroeconomic data covering a long historical period from January 1960 to December 2025, sourced from the Federal Reserve Bank of St. Louis database (FRED). The three series under study are inflation, real GDP, and the short-term interest rate.

Price index and inflation. The price index is the Consumer Price Index for All Urban Consumers:

All Items in U.S. City Average (CPIAUCSL), observed at a monthly frequency and seasonally adjusted by the Bureau of Labor Statistics (Federal Reserve Bank of St. Louis, n.d.b). The series runs from January 1960 to December 2025 (792 monthly observations), with one missing value in October 2025 that is dropped. Inflation is constructed as the monthly log-difference of the price index, scaled to percentage points:

$$\pi_t = 100\big[\ln(\mathrm{CPI}_t) - \ln(\mathrm{CPI}_{t-1})\big].$$

The resulting inflation series covers February 1960 to December 2025 (790 monthly observations).

Real Gross Domestic Product. GDP is measured by the FRED series GDPC1, which reports real GDP in billions of chained 2017 dollars at a seasonally adjusted annual rate (Federal Reserve Bank of St. Louis, n.d.f). The series is available at a quarterly frequency and covers 1960Q1–2025Q4 (264 quarterly observations). As in Problem I, we work with the logarithm of real GDP, $\log(\mathrm{GDP}_t)$, so that first differences yield the annualised quarterly growth rate $400\,\Delta\log(\mathrm{GDP}_t)$.

Short-term interest rate. The short-term interest rate is proxied by the Federal Funds Effective Rate (FEDFUNDS), observed at a monthly frequency and reported in percent per annum (Federal Reserve Bank of St. Louis, n.d.d). The series is not seasonally adjusted and covers January 1960 to December 2025 (792 monthly observations).

Table 9 summarises the key features of each series.

Table 9: Data summary for Problem II

Series	Source	Frequency	Sample	T
$\log(\mathrm{CPI}_t)$	FRED CPIAUCSL	Monthly	Feb 1960 – Dec 2025	790
π_t (inflation)	derived from CPI	Monthly	Feb 1960 – Dec 2025	790
$\log(\mathrm{GDP}_t)$	FRED GDPC1	Quarterly	1960Q1 – 2025Q4	264
r_t (fed funds rate)	FRED FEDFUNDS	Monthly	Jan 1960 – Dec 2025	792

2.1 Best ARMA model for the 3 variables (including all the steps)

We follow the six steps reviewing what we did in class to find the best ARMA model for each of the three variables: inflation, GDP growth, and the Federal Funds rate. All estimation is done in R using maximum likelihood.

The first step consists on data construction. The three series are constructed from the FRED downloads described in Table 9. We work with the following transformations:

- **Inflation**: the annualized monthly log-change of the CPI, $\pi_t = 1200 \times \Delta \ln \mathrm{CPI}_t$. Multiplying by 1200 converts the monthly log-difference into an annualised percentage.
- **GDP growth**: we first take the logarithm of real GDP, $\ln \mathrm{GDP}_t$, and then compute the annualised quarterly growth rate $g_t = 400 \times \Delta \ln \mathrm{GDP}_t$. Multiplying by 400 converts the quarterly log-difference

into an annualised percentage.

- **Federal Funds rate**: we start with the level of the effective federal funds rate r_t and decide whether to model it in levels or first differences based on the ADF test we will explore in the following steps.

Figure 26 plots the three series over the full sample. We observe, that inflation fluctuates around a positive mean with no clear deterministic trend, though it displays substantial swings, most notably the sharp disinflation of the early 1980s and the post-pandemic surge of 2021–2022. Log real GDP shows a clear and persistent upward trend throughout the sample, punctuated by recessions that appear as brief dips; it never reverts to a fixed mean, which is the visual hallmark of a non-stationary, $I(1)$ process. The Federal Funds rate moves in long policy-driven cycles, rising sharply in the Volcker era, declining through the 2000s, falling to the zero lower bound after 2008, and then rising again from 2022 onward. Its slow-moving nature makes it difficult to classify visually as either stationary or $I(1)$, which is why we rely on the formal ADF test below.

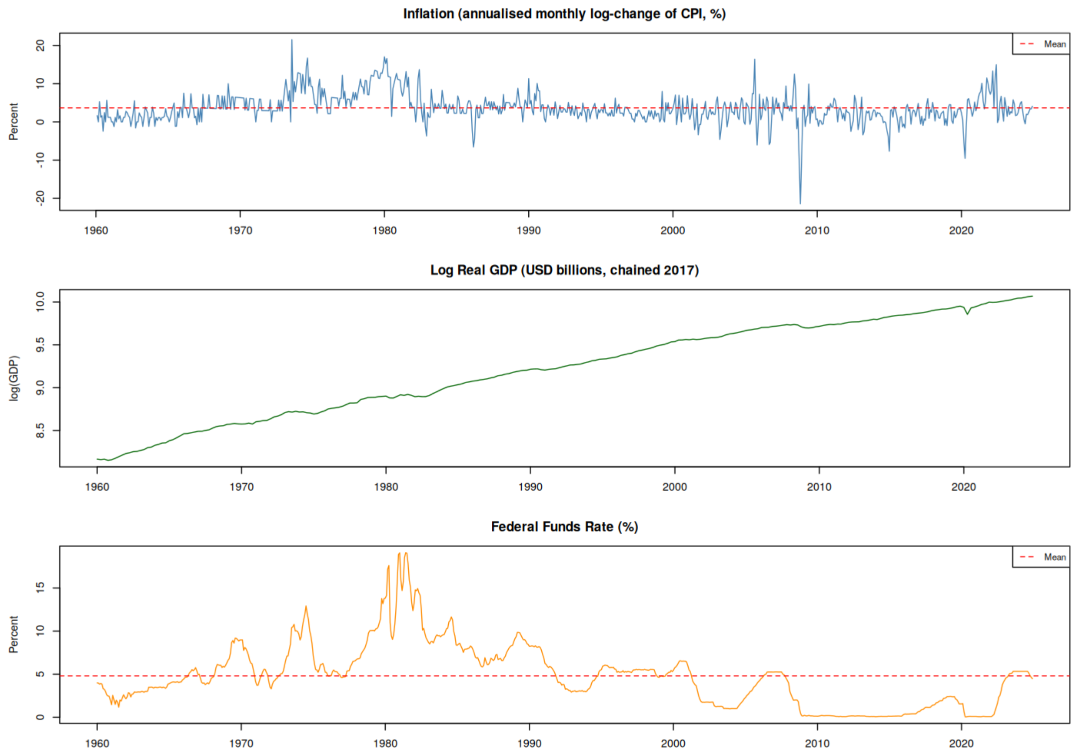

Figure 26: Raw series used in Problem II. Top: annualised inflation $\pi_t = 1200\,\Delta \ln \mathrm{CPI}_t$ (%). Middle: log real GDP. Bottom: Federal Funds rate (%). The red dashed line shows the sample mean for inflation and the Fed Funds rate. Sample: 1960Q1–2024Q4 (quarterly GDP) and January 1960–December 2024 (monthly series).

Before fitting any ARMA model, we must verify that each series is covariance-stationary. To verify this, we apply the Augmented Dickey–Fuller (ADF) test, selecting lag length by AIC and choosing the deterministic specification (drift or trend) based on the visual inspection of Figure 26.

Table 10: ADF unit-root test results. H_0: the series has a unit root. Lag lengths are selected by AIC. The specification "drift" includes a constant; "trend" includes a constant and a linear time trend. Critical values at 1%, 5%, and 10% are shown.

Series	Spec.	Lags	$\hat{\tau}$	CV 1%	CV 5%	CV 10%
Inflation	drift	1	-10.887	-3.43	-2.86	-2.57
Log real GDP	trend	1	-2.025	-3.98	-3.42	-3.13
GDP growth	drift	1	-9.983	-3.44	-2.87	-2.57
Fed Funds (level)	drift	1	-2.906	-3.43	-2.86	-2.57
Fed Funds (Δ)	drift	1	-18.092	-3.43	-2.86	-2.57

We proceed now to discuss the result for each series in turn.

Inflation. The test statistic is $\hat{\tau} = -10.89$, which is far below the 5% critical value of -2.86. We strongly reject the unit root null at all conventional significance levels. Inflation is stationary in levels and can be modelled directly with an ARMA model without any further transformation.

Log real GDP. We use the "trend" specification because the plot in Figure 26 shows a clear upward trend. The test statistic is $\hat{\tau} = -2.03$, which is well above the 5% critical value of -3.42. We fail to reject the unit root null. Log GDP is $I(1)$, so we take its first difference to obtain the annualised GDP growth rate $g_t 400\,\Delta\ln\mathrm{GDP}_t$. The ADF test on g_t gives $\hat{\tau} = -9.98$, decisively rejecting the unit root. GDP growth is stationary and is the series we bring to the ARMA step.

Federal Funds rate. The result here is more nuanced. The test statistic in levels is $\hat{\tau} = -2.91$, which is just below the 5% critical value of -2.86, so we reject the unit root at the 5% level but not at the 1% level ($-2.91 > -3.43$). The evidence is therefore borderline. The first difference yields $\hat{\tau} = -18.09$, which clearly rejects the unit root, confirming the series is at most $I(1)$. Given the borderline nature of the level test, we proceed conservatively by keeping the Fed Funds rate in levels as our baseline stationary series, noting that the ADF test does reject at the conventional 5% threshold.

So, the three stationary series we carry forward to the ARMA model selection step are: **inflation** π_t, **GDP growth** g_t, and **the Fed Funds rate in levels** r_t.

Before running the formal grid search, we inspect the sample ACF and PACF of each stationary series to get an intuition for the type and order of an ARMA model that may be appropriate. The results are shown in Figure 27.

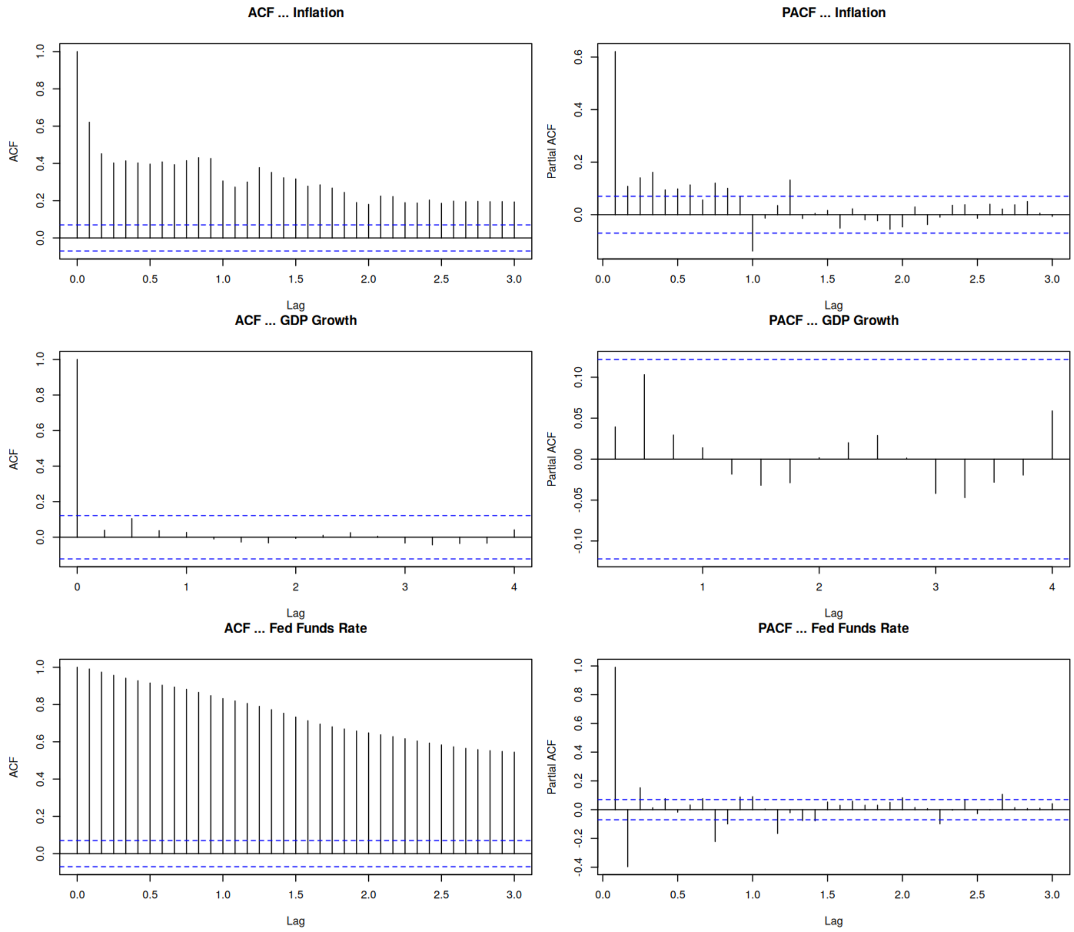

Figure 27: Sample ACF (left) and PACF (right) for the three stationary series. Top row: inflation. Middle row: GDP growth. Bottom row: Federal Funds rate. Dashed lines are 95% confidence bands under the white-noise null hypothesis.

Inflation. The ACF decays slowly and remains positive for many lags, suggesting strong persistence. The PACF shows a dominant spike at lag 1 followed by smaller but non-negligible values at higher lags. This pattern is consistent with a highly persistent AR-type process. We expect the best model to have a meaningful autoregressive component, though the order is not obvious from the graph alone.

GDP growth. Both the ACF and the PACF are very close to zero after lag 1, with most values comfortably inside the confidence bands. This suggests that GDP growth has very weak serial dependence and may be close to white noise. A low-order model, or even ARMA(0,0), seems plausible.

Federal Funds rate. The ACF decays extremely slowly and remains large and positive for many lags — the classic signature of a highly persistent process. The PACF shows a single dominant spike at lag 1 and much smaller values afterwards, consistent with a strong AR(1)-type structure, though higher-order AR dynamics may also be present given the slow ACF decay.

We use these plots only as guidance. The ACF and PACF are informative but not sufficient to pin down the exact model order, particularly for persistent series. We therefore proceed to a systematic grid search in the following steps.

Therefore, now we estimate ARMA(p, q) models by maximum likelihood for a range of orders and rank them by the Bayesian Information Criterion (BIC). We prefer BIC over AIC because BIC penalises model complexity more strongly and is consistent for the true order in large samples. However, we also report AIC as a secondary check.

A necessary condition for accepting a model is that its residuals look like white noise. We verify this using the Box–Pierce test with $H = 24$ lags for monthly series and $H = 12$ lags for quarterly series. A model is acceptable only if its Box–Pierce p-value exceeds 0.05, meaning we do not reject the white-noise null for the residuals. Our selection rule is therefore: among all models that pass the Box–Pierce test, we choose the one with the lowest BIC.

GDP growth. We search over ARMA(p, q) with $p, q \in \{0, \ldots, 3\}$. The results are shown in Table 11.

Table 11: ARMA grid search for GDP growth ($p, q \in \{0, \ldots, 3\}$), sorted by BIC. The Box–Pierce test uses $H = 12$ lags. All models shown pass the residual white-noise test.

p	q	Log-lik	AIC	BIC	BP p-val
0	**0**	−741.04	1486.08	**1493.19**	0.966
1	0	−740.84	1487.67	1498.34	0.969
0	1	−740.87	1487.74	1498.41	0.966
2	0	−739.45	1486.90	1501.13	0.999
0	2	−739.57	1487.14	1501.37	0.999
1	1	−740.05	1488.09	1502.32	0.990
1	2	−739.31	1488.61	1506.39	0.999
2	1	−739.33	1488.66	1506.44	0.999
3	0	−739.33	1488.67	1506.45	0.999
0	3	−739.39	1488.78	1506.56	0.998

The ARMA$(0, 0)$ model is a white-noise process around a constant mean that achieves the lowest BIC of all candidates and passes the Box–Pierce test with $p = 0.97$. Adding any AR or MA terms increases BIC without meaningfully improving the fit. This is fully consistent with the flat ACF and PACF observed in Figure 27: GDP growth over this long sample is essentially unpredictable from its own past. Therefore, **we select ARMA$(0, 0)$ for GDP growth.**

Inflation. We begin with an ARMA grid search over $p, q \in \{0, \ldots, 4\}$. The models with the lowest BIC are low-order specifications such as ARMA$(1, 2)$ and ARMA$(2, 1)$, but all of them have Box–Pierce p-values extremely close to zero, meaning we strongly reject the white-noise null for their residuals. The same result holds when we expand the grid to $p, q \in \{0, \ldots, 6\}$.

We therefore switch to a pure AR search, which is more parsimonious for highly persistent series and avoids the numerical convergence problems that sometimes arise with mixed ARMA models. We estimate AR(p) models for $p = 0, 1, \ldots, 12$. None of them passes the Box–Pierce test: even the best BIC model, AR(12), has $p = 0.011$. Since AR(12) sits exactly at the boundary of the search grid, we extend the search to $p = 24$.

Table 12: Pure AR grid search for inflation ($p \in \{0, \ldots, 24\}$), top 15 models sorted by BIC. The Box–Pierce test uses $H = 24$ lags. ✓ marks models that pass (p-val > 0.05).

p	Log-lik	AIC	BIC	BP p-val
12	−1892.17	3812.34	3877.55	0.011
15	**−1884.58**	**3803.15**	**3882.34**	**0.391** ✓
10	−1901.63	3827.26	3883.15	< 0.001
13	−1892.12	3814.24	3884.11	0.006
9	−1905.59	3833.17	3884.41	< 0.001
7	−1912.50	3843.00	3884.92	< 0.001
11	−1899.76	3825.52	3886.08	< 0.001
6	−1917.53	3851.05	3888.32	< 0.001
16	−1884.49	3804.99	3888.83	0.303 ✓
8	−1911.23	3842.45	3889.03	< 0.001
5	−1921.38	3856.76	3889.37	< 0.001
4	−1924.85	3861.69	3889.64	< 0.001
14	−1891.66	3815.32	3889.85	0.005
17	−1884.47	3806.95	3895.45	0.221 ✓
18	−1884.39	3808.77	3901.93	0.144 ✓

Among all AR models that pass the Box–Pierce test, AR(15) has the lowest BIC of 3882.34 and a comfortable p-value of 0.39. The next acceptable models are AR(16) and AR(17), both with higher BIC. The need for 15 lags is striking but not surprising: over a sample that spans 65 years and includes the Great Inflation of the 1970s, the Volcker disinflation, and the post-pandemic surge, inflation exhibits very long-run persistence that only a high-order AR model can capture. Thus, we end up **selecting AR**(15) **for inflation.**

Federal Funds rate. The ACF and PACF in Figure 27 strongly suggested a highly persistent AR process. We therefore go directly to a pure AR search over $p = 0, 1, \ldots, 24$.

Table 13: Pure AR grid search for the Federal Funds rate ($p \in \{0, \ldots, 24\}$), top 10 models sorted by BIC. The Box–Pierce test uses $H = 24$ lags. ✓ marks models that pass (p-val > 0.05).

p	Log-lik	AIC	BIC	BP p-val
14	−439.03	910.06	984.61	0.014
16	−436.52	909.03	992.90	0.034
17	**−433.91**	**905.82**	**994.35**	**0.095** ✓
10	−457.67	939.34	995.25	< 0.001
12	−451.27	930.54	995.77	< 0.001
11	−454.61	935.23	995.80	< 0.001
9	−461.42	944.83	996.08	< 0.001
18	−432.42	904.84	998.02	0.172 ✓
13	−451.26	932.51	1002.40	< 0.001
19	−432.08	906.15	1004.00	0.116 ✓

Lower-order AR models all fail the Box–Pierce test, with p-values essentially at zero. The first model to pass is AR(17), with a Box–Pierce p-value of 0.095 and the lowest BIC among the acceptable models. AR(14) and AR(16) have lower BIC values but fail the residual diagnostic, so they are not acceptable. The high order required again reflects the extreme persistence of the Fed Funds rate: monetary policy changes unfold slowly over many months, and a long AR lag structure is needed to capture that behaviour. So, **we select AR(17) for the Federal Funds rate.**

Finally, now estimate the three selected models and run a full battery of diagnostic checks: (i) the Box–Pierce test on residuals, (ii) the Jarque–Bera normality test, (iii) a visual inspection of residuals, ACF, PACF, and QQ-plot, and (iv) the stationarity and invertibility check via the roots of the AR and MA polynomials.

Table 14 summarises the main numerical results.

Table 14: Summary diagnostics for the three final ARMA models. The Box–Pierce test uses $H = 24$ (monthly) and $H = 12$ (quarterly) lags. The Jarque–Bera test has the null of Gaussian residuals.

Series	Model	Log-lik	AIC	BIC	BP stat	BP p-val	JB stat	JB p-val
GDP Growth	ARMA(0, 0)	−741.04	1486.08	1493.19	4.75	0.966	6643.9	< 0.001
Inflation	AR(15)	−1884.58	3803.15	3882.34	9.52	0.391	1404.2	< 0.001
Fed Funds	AR(17)	−433.91	905.82	994.35	12.18	0.095	103345.7	< 0.001

Box–Pierce test. All three models pass: the residuals cannot be distinguished from white noise at the 5% level. This is the key diagnostic we require from an acceptable model.

Jarque–Bera test. Normality is rejected for all three series. This is common with macroeconomic data covering long samples: the 1973 oil shock, the 1987 crash, the 2008 financial crisis, the 2020 Covid contraction, and the 2021–2022 inflation episode all generate outliers that produce fat tails in the residual distribution. Rejection of normality does not invalidate the ARMA specification; it simply means that prediction intervals based on the Gaussian assumption would be too narrow.

Stationarity and invertibility. For both AR(15) (inflation) and AR(17) (Fed Funds), we verify that all roots of the characteristic AR polynomial lie strictly outside the unit circle (modulus > 1), confirming that the processes are covariance-stationary. There are no MA components, so invertibility is satisfied automatically.

Residual plots. Figures 28, 29, and 30 show the residual time series, ACF, PACF, and QQ-plot for each model.

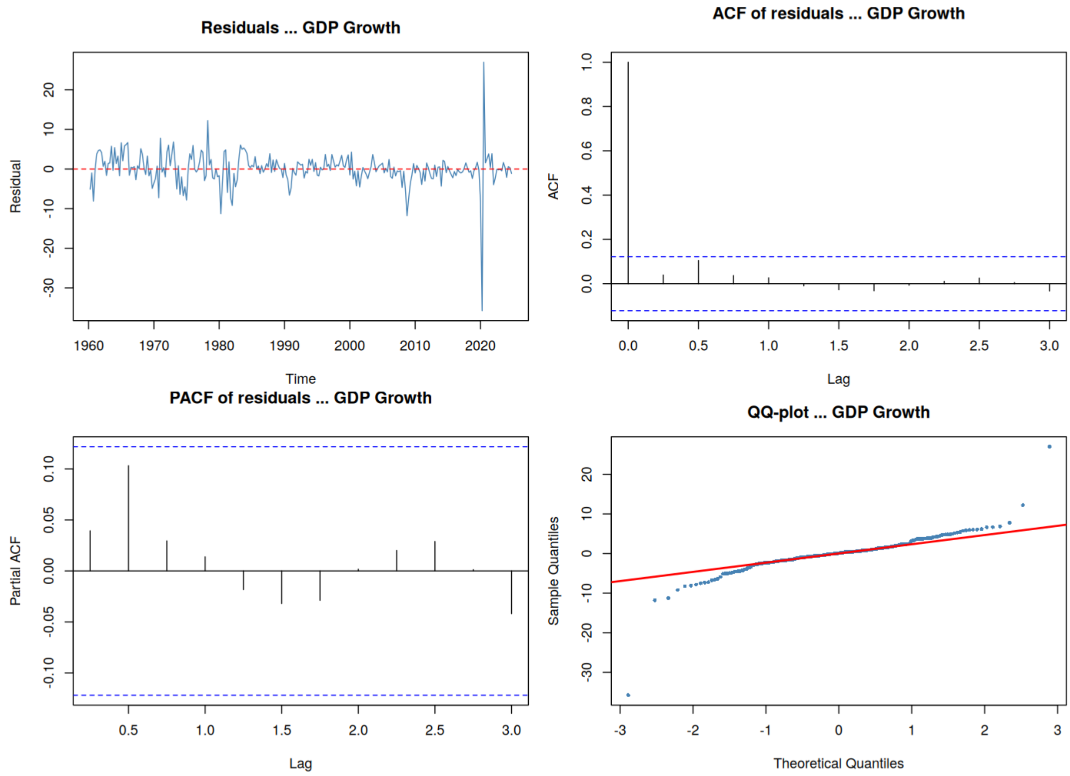

Figure 28: Residual diagnostics for GDP growth — ARMA$(0, 0)$. The residual ACF and PACF show no significant spikes, confirming that the model adequately captures all serial dependence. The QQ-plot shows heavy tails driven by the Covid quarters.

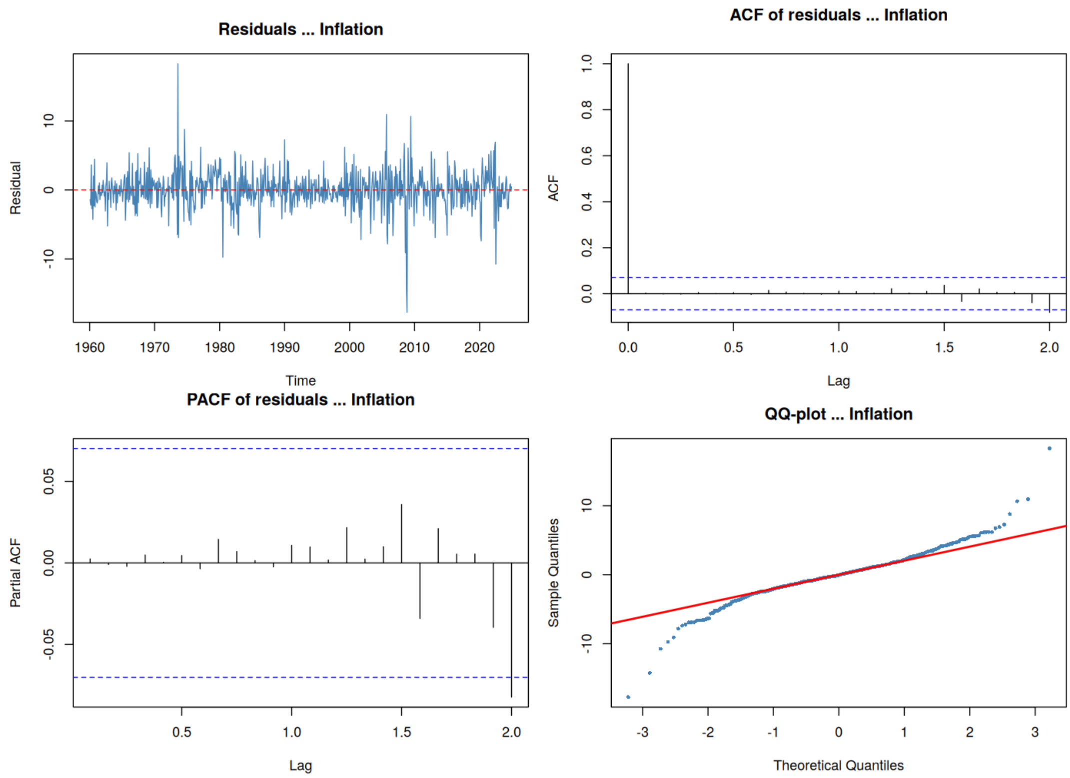

Figure 29: Residual diagnostics for inflation — AR(15). The residual ACF and PACF are clean, with all spikes inside the 95% confidence bands. The QQ-plot reveals fat tails from episodes such as the 1973–1974 oil shock and the 2021–2022 inflation surge.

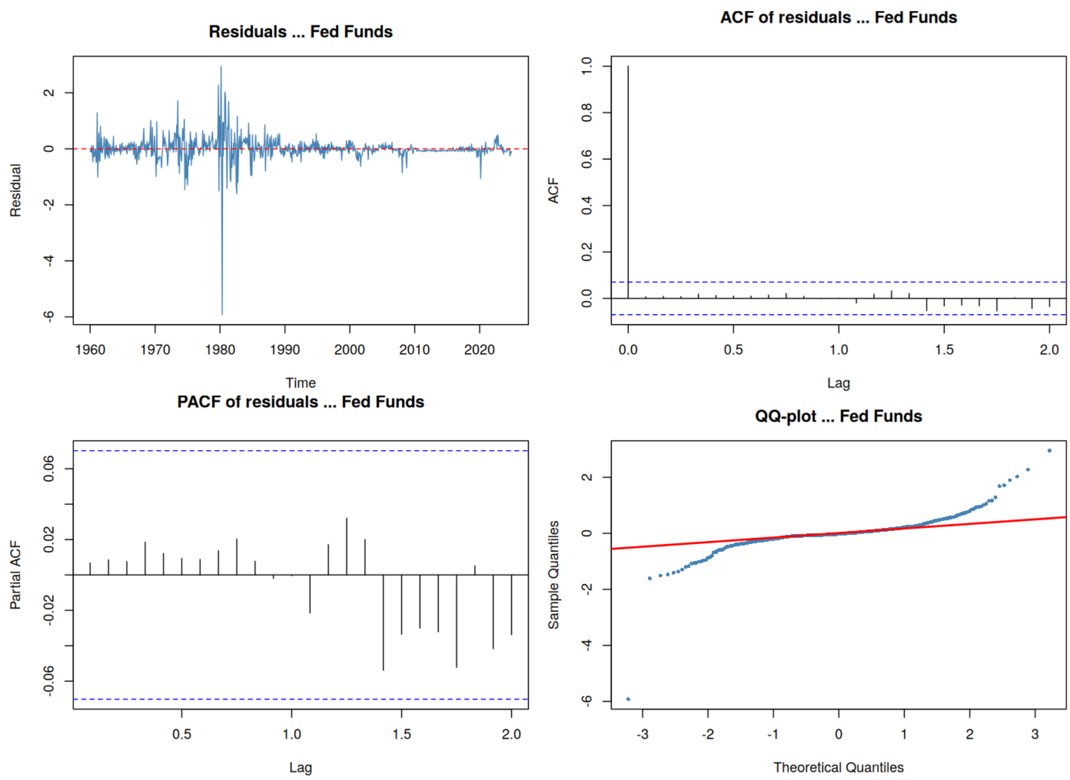

Figure 30: Residual diagnostics for the Federal Funds rate — AR(17). The residual ACF and PACF are clean. The QQ-plot shows extremely heavy tails, reflecting the large discrete jumps associated with FOMC decisions such as the emergency cuts of 2008 and 2020.

Summarizing, Table 15 collects the final model selection for all three variables.

Table 15: Final ARMA models selected for Problem II, Question 1.

Variable	Selected model	BIC	Justification
GDP growth	ARMA$(0, 0)$	1493.19	Lowest BIC; BP $p = 0.97$
Inflation	AR(15)	3882.34	Lowest BIC passing BP; $p = 0.39$
Fed Funds	AR(17)	994.35	Lowest BIC passing BP; $p = 0.095$

The results paint a coherent economic picture. GDP growth over this long sample is essentially white noise: once we control for its unconditional mean, past growth rates contain no information about future growth. Inflation, by contrast, is highly persistent and requires 15 lags to adequately whiten its residuals, a consequence of the long inflation cycles visible in the data since the 1960s. The Federal Funds rate is the most persistent of the three variables: 17 autoregressive lags are needed because monetary policy changes unfold slowly and in a highly path-dependent way. Both the inflation and Fed Funds models reject the normality of residuals, reflecting the presence of large structural shocks over this 65-year sample that no linear Gaussian model can fully capture.

2.2 In sample and out-of-sample forecasting comparison

The goal of this question is to study the forecasting in sample and out-of-sample using different models for each of the three variables used in question 1:

- Inflation π_t
- GDP growth g_t
- Federal Funds rate in level r_t

Model selection For each of the variables we will take three different models to compare their forecast properties. First of all, we will select the three ARMA(p,q) models from question 1 of problem 2 as we already seen that there are the best ARMA models. In other words, we select:

- ARMA(0,0) = White noise, for GDP growth
- ARMA(15,0) = AR(15) for inflation
- ARMA(17,0) = AR(17) for Federal Funds rate

GDP growth presents no autocorrelation (every spikes at lag larger than 0 are within the confidence interval), the white noise results of an ARMA(0,0) is very plausible. Therefore, we will compare the forecasts of this model with an AR(1) and an ARMA(1,1). According to the time series properties of GDP growth, testing more complex models will not lead to better forecasts.

The inflation series presents a sharp declines of PACF at lag 1, even though some other lags are significant. In the same time, the ACF slowly decays from lag 1 onward, for these reasons we will use an AR(1) and an ARMA(1,1) for forecast comparison.

Finally, for the Federal Funds rate series, the ACF slowly decreases with a PACF significant at lag 1 followed by a sharp declines with few significant lags after. Again, we will test an AR(1) and a ARMA(1,1). All the models are presented in Table 16.

Variable	Model 1	Model 2	Model 3
GDP growth	ARMA(0,0)	AR(1)	ARMA(1,1)
Inflation	AR(15)	AR(1)	ARMA(1,1)
Federal Funds Rate	AR(17)	AR(1)	ARMA(1,1)

Table 16: Models considered for forecasting comparison

GDP growth

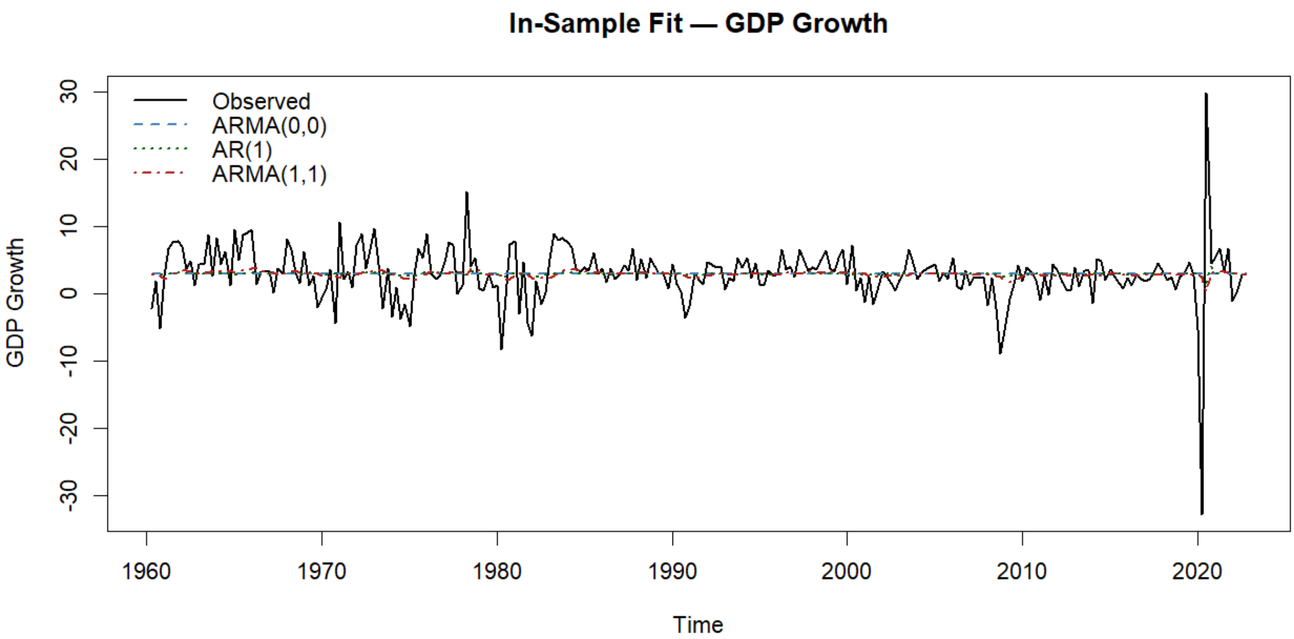

Figure 31: In sample fit: GDP growth

Model	AIC	BIC	RMSE (train)	MAE (train)
ARMA(0,0)	1447.67	1454.73	4.2927	2.5660
AR(1)	1449.27	1459.85	4.2892	2.5353
ARMA(1,1)	1449.72	1463.82	4.2759	2.4962

Table 17: In-sample performance: GDP growth

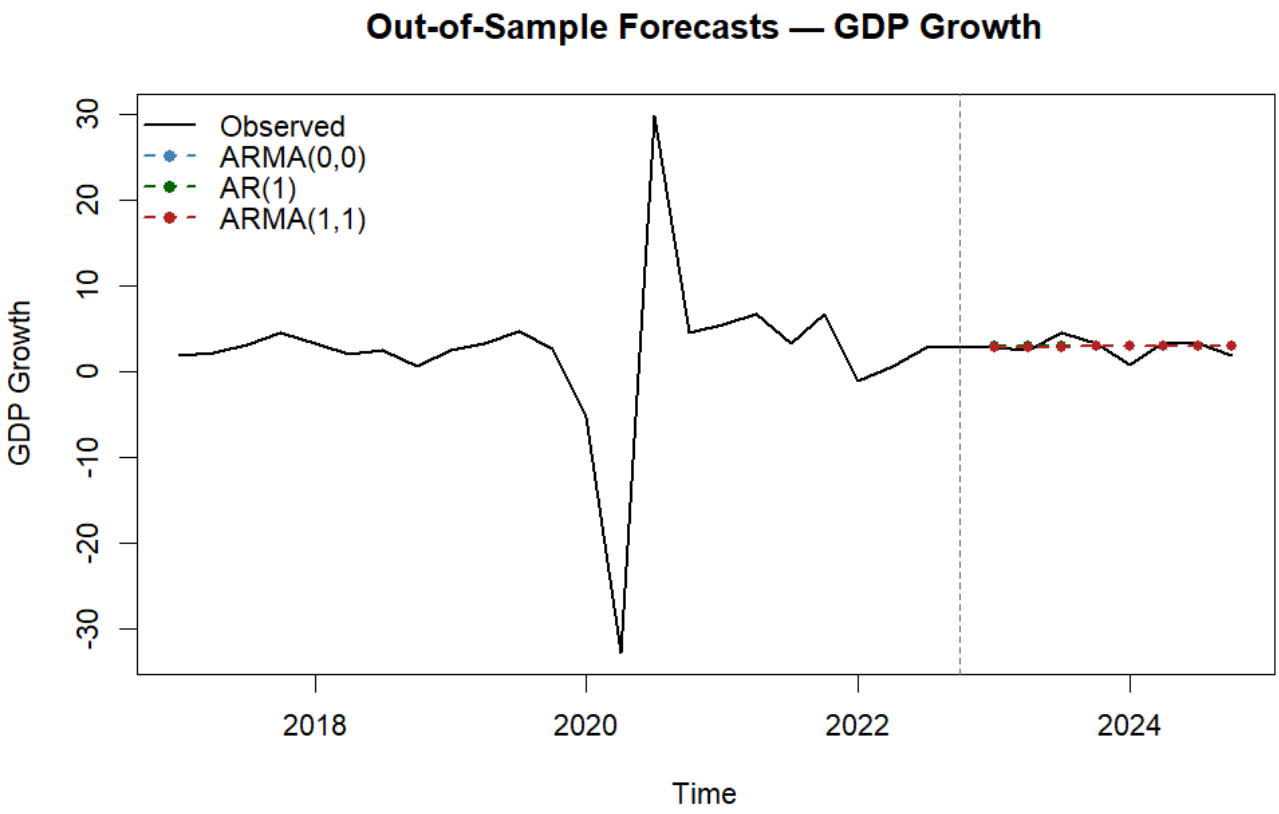

Figure 32: Out of sample forecasts: GDP growth

Model	RMSE (test)	MAE (test)
ARMA(0,0)	1.0718	0.8374
AR(1)	1.0716	0.8365
ARMA(1,1)	1.0730	0.8341

Table 18: Out-of-sample forecast performance: GDP growth (h = 8)

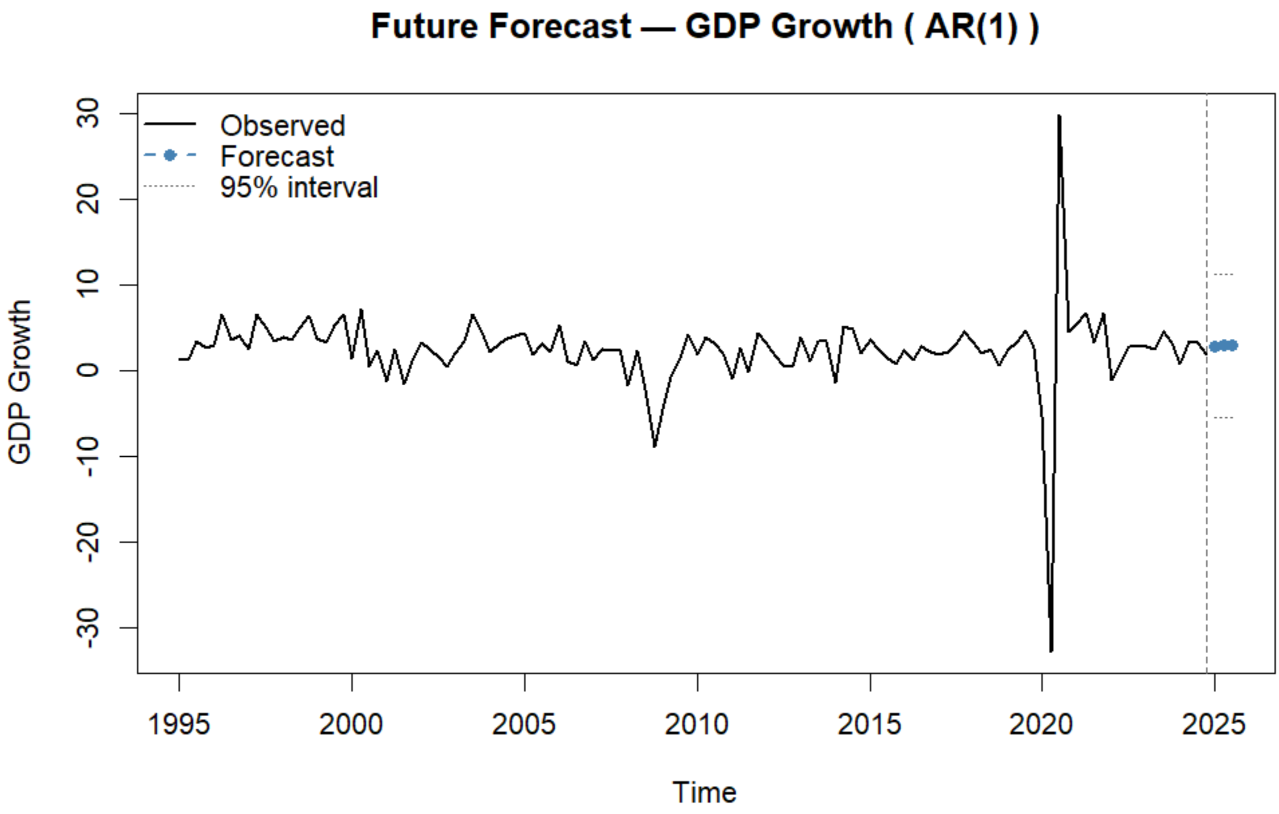

Figure 33: Caption

Period	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025 Q1	2.8946	-2.5433	8.3324	-5.4219	11.2110
2025 Q2	2.9363	-2.5057	8.3784	-5.3866	11.2592
2025 Q3	2.9380	-2.5041	8.3800	-5.3849	11.2609

Table 19: Future forecasts for GDP growth using the selected AR(1) model

For GDP growth the sample size is divided into a training set, from 1960 Q1 to 2022 Q4, and a test set, from 2023 Q1 to 2024 Q4. Therefore, for out-of-sample forecast we select $h = 8$ to have a 2-year ahead forecast. In addition, we choose to perform 3 step ahead of future forecasts (above our observed data), as GDP growth is in a quarterly format this corresponds to 3 quarters equals 9 months.

Table 17 and Figure 31 illustrate the results obtained for the In-sample fitting. First, in Figure 31, we can observe that all the 3 models predicts poorly the fit of the GDP growth as the 3 dotted lines (one for each models) do not align well with the observed fit. The predicted fits look like to correspond to the mean of the series.

Moreover, in Table 17 we can notice that the best model according to AIC and BIC criterion is still the ARMA(0,0) model while RMSE and MAE indicates other results. One remark is that AIC and BIC for ARMA(0,0) are different than those obtained in question 1, this is due to the fact that in question 2 we calculate both of them on the training set, while in question 1 that was on the entire sample. The ARMA(1,1) model is the one with the lowest RMSE and MAE (4.2759 and 2.4962) , this indicates that its forecasts are the closest to the observed data. By contrast, the "supposed" best model, ARMA(0,0), is the one with the highest metrics (RMSE = 4.2927 and MAE = 2.5660). To put it in perspective, even if we notice differences, there are very small between the three models, according to metrics and graphs we can state that all the models give poor in-sample forecasts.

The out-of-sample performance can be graphically observed in Figure 32. The forecasts of the three models are aligned and correspond to a straight line: the mean of the series. The models do not succeed to replicate the fluctuations observed in the data. Table 17 illustrates the forecasts accuracy for the three models. We notice that the best one is the AR(1) model but all RMSE and MAE values are very close. The results observed for the out-of-sample forecasts are similar than those observed for the in-sample ones: models predict poorly g_t. This was expected as the series is similar to a white noise process which indicates that past values do not help to predict future values. The forecasts values correspond to the mean of the series, which are obviously not good estimates.

This is again what we observed for future forecasts of g_t. We selected the AR(1) as it is the one which displays the lowest RMSE and MAE values for out-of-sample forecasts. Figure 33 shows the next three periods predicted, we can see that all are quite similar and seems to correspond to the mean of the series. By looking at Table 19, the values of the three forecasts are 2.8946 for 2025 Q1, 2.9363 for 2025 Q2 and 2.9380 for 2025 Q3. In the same time, the mean of the series is:

$$\bar{g} = \frac{1}{259} \sum_{t=1}^{250} g_t = 2.939022$$

In addition the confidence intervals at 80% and 95% are very large, indicating poor forecasts.

We can conclude that as the GDP growth series exhibits very weak autocorrelation, as shown by the ACF and PACF plots. This suggests that the series behaves close to a white noise process. As a result, past values contain little information about future observations, making accurate forecasting difficult. Consequently, all models produce very similar forecast performance and values close to the empirical mean of the series.

Inflation

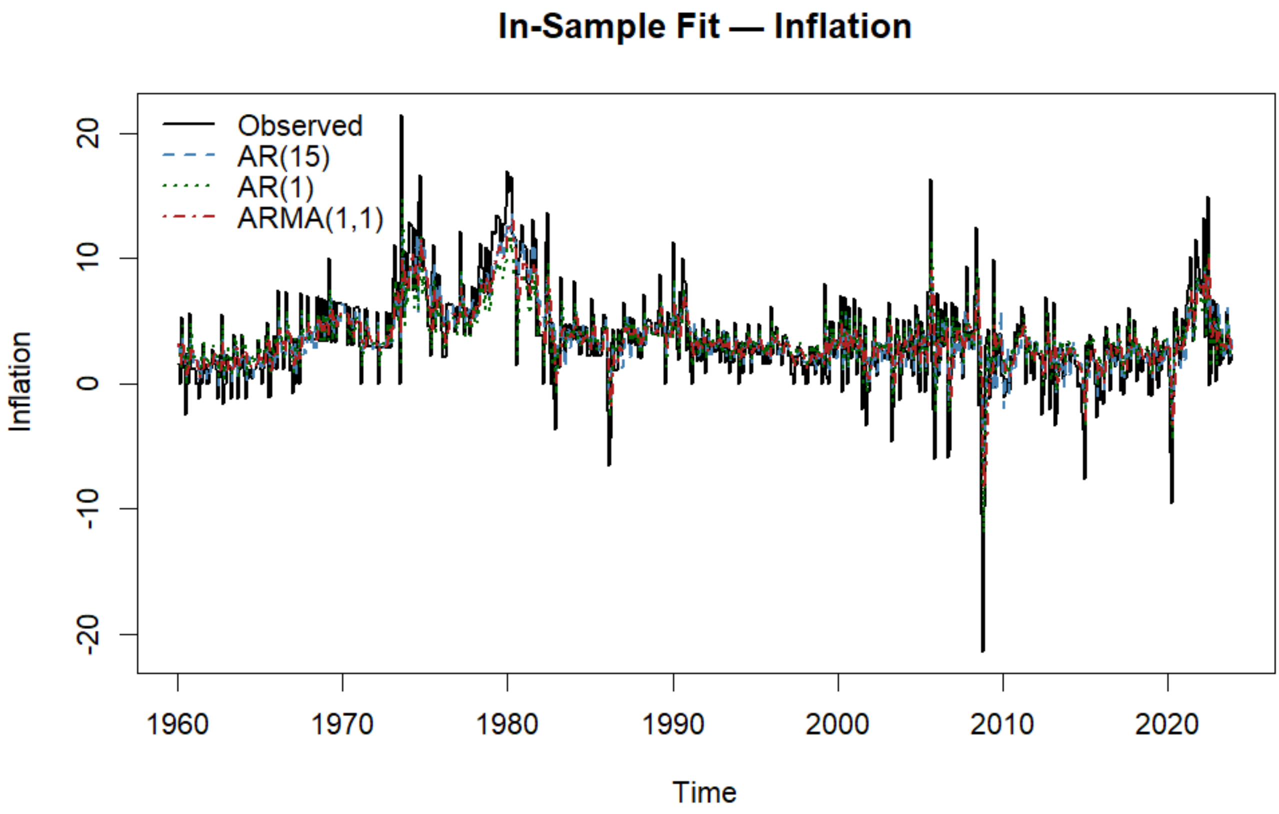

Figure 34: In sample fit: Inflation

Model	AIC	BIC	RMSE (train)	MAE (train)
AR(15)	3753.59	3832.51	2.7311	1.9190
AR(1)	3850.38	3864.30	2.9649	2.1154
ARMA(1,1)	3828.97	3847.54	2.9197	2.0415

Table 20: In-sample performance of candidate models for inflation

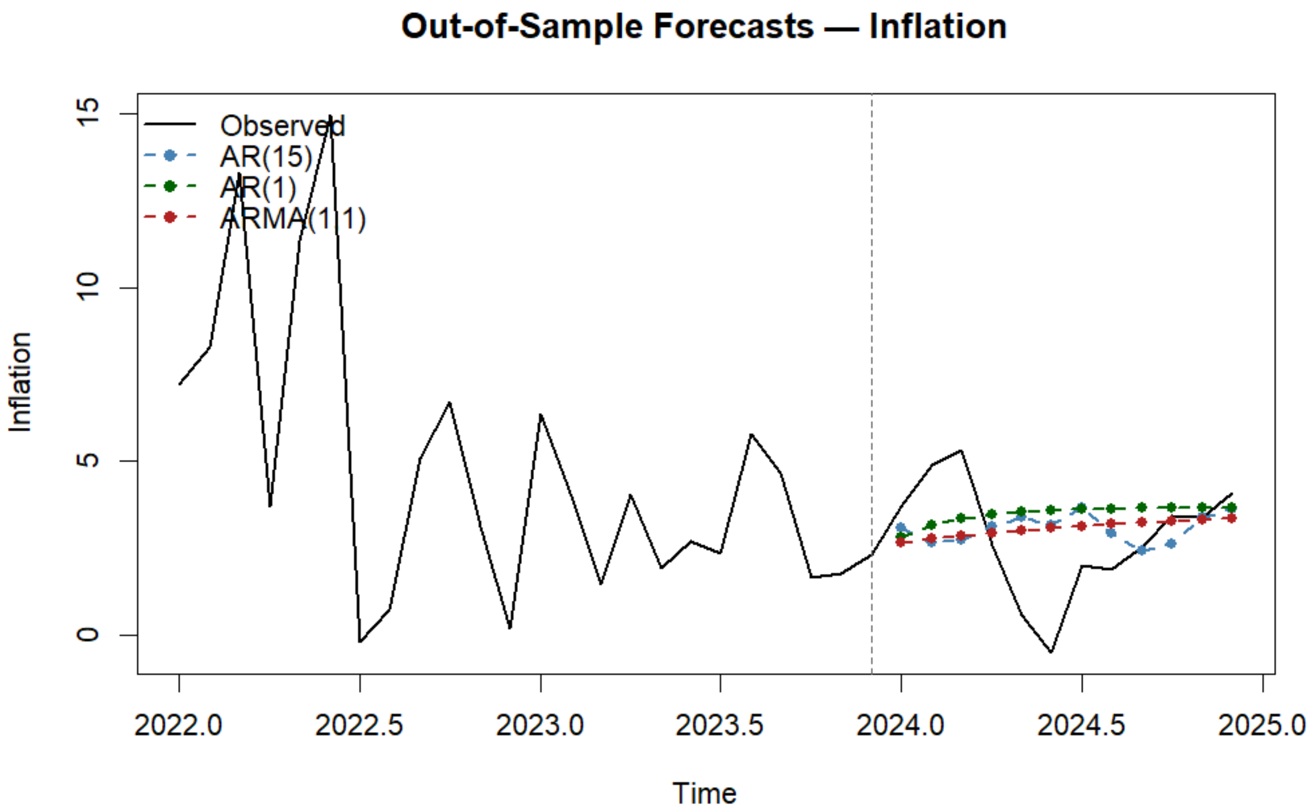

Figure 35: Out of sample forecasts: Inflation

Model	RMSE (test)	MAE (test)
AR(15)	1.7866	1.3776
AR(1)	1.8546	1.4932
ARMA(1,1)	1.6941	1.3361

Table 21: Out-of-sample forecast accuracy for inflation (h = 12)

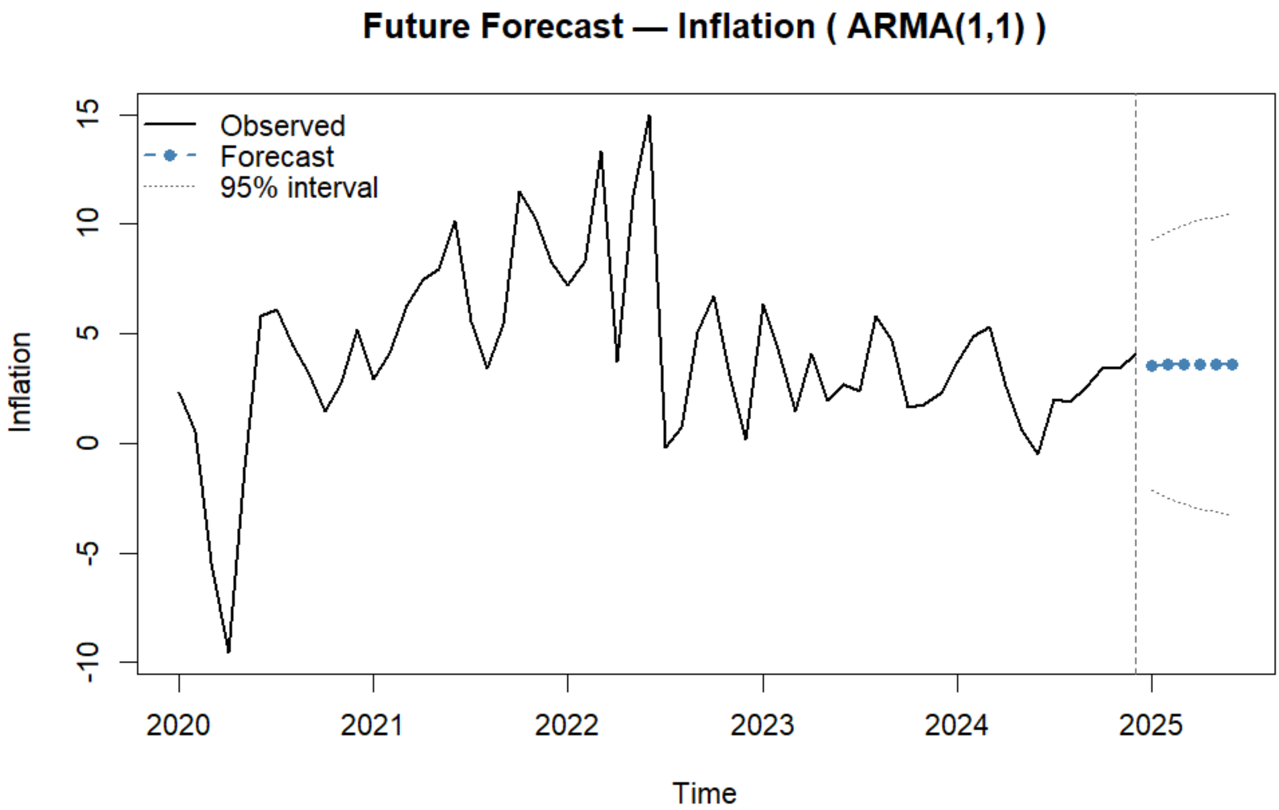

Figure 36: Future forecasts: Inflation

Period	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2025	3.5671	-0.1616	7.2957	-2.1355	9.2696
Feb 2025	3.5765	-0.4005	7.5535	-2.5058	9.6588
Mar 2025	3.5849	-0.5788	7.7486	-2.7829	9.9527
Apr 2025	3.5924	-0.7138	7.8986	-2.9933	10.1782
May 2025	3.5991	-0.8170	8.0152	-3.1548	10.3530
Jun 2025	3.6051	-0.8964	8.1065	-3.2794	10.4895

Table 22: Future forecasts for inflation using the ARMA(1,1) model

The training set of the inflation series is from January 1960 to December 2023, we keep a test of 12 monthly observations which correspond to 1 year (2024). Therefore we will have 1 year ahead of out-of-sample forecast. In addition, we choose to perform a 6 steps ahead future forecasts (above our observed data), which correspond to the first 6 months of 2025.

Figure 34 and Table 20 illustrate the results obtained for the In sample forecasts. We can observe in Figure 34 that the three models gives quite similar performance by fitting the data. All the three follows the general trend of the data but dot not fit well when there is huge fluctuations (peaks or troughs). Moreover, in Table 20, the results show that the best model, according to information criterion, for the in sample performance is the ARMA(1,1). Nevertheless, according to forecast accuracy metrics, namely RMSE and MAE, the best model is indeed the AR(15) with the lowest values. These results are consistent with those found in question 1.

We made out-of-sample forecasts on the entire 2024 year. We can observe in Figure 35 that the three models predicts poorly the dynamic of inflation. We have the mean of the inflation series:

$$\bar{\pi} = \frac{1}{779} \sum_{t=1}^{779} \pi_t = 3.66752$$

AR(1) and ARMA(1,1) give predictions around the mean of the series while the AR(15) model "succeed" to forecast a a somehow trend of inflation, even if it remains far from the observed one. By checking RMSE and MAE values in Table 21 we can see that the best model is not the AR(15) one. Indeed, the RMSE values are 1.7866 for AR(15), 1.8546 for AR(1) and 1.6941 for ARMA(1,1). The ARMA model is the one with best accuracy for predicting inflation.

For future forecasts we used the ARMA(1,1) model. The results are presented in Figure 36 and Table 22. Both illustrate that the forecasts from January to June 2025 are around the empirical mean of the series indicating poor predicting performance. As for g_t, the confidence interval are very wide confirming uncertainty.

Nevertheless, our empirical study stop in 2024 but in reality we observe data for the year 2025, therefore we can compare our future forecasts with reality. Indeed, in 2025, the average inflation rate in USA was 2.6%. Respectively, 3%, 2.8% and 2.4% in January, February and March 2025. Our forecasts are therefore far away from reality.

To wrap up , the results highlight a clear difference between in-sample fit and out-of-sample forecasting performance. While the AR(15) model provides the best in-sample fit according to RMSE and MAE, it performs relatively poorly for out-of-sample forecasts. This suggests that the AR(15) model may suffer from overfitting. In contrast, the ARMA(1,1) model gives the best out-of-sample performance. Therefore, while the ARMA(1,1) model provides the best forecasting performance among the candidates, its predictive power remains limited. This suggests that inflation dynamics are only partially captured by simple ARMA model, especially during periods of large fluctuations.

Federal Funds rate

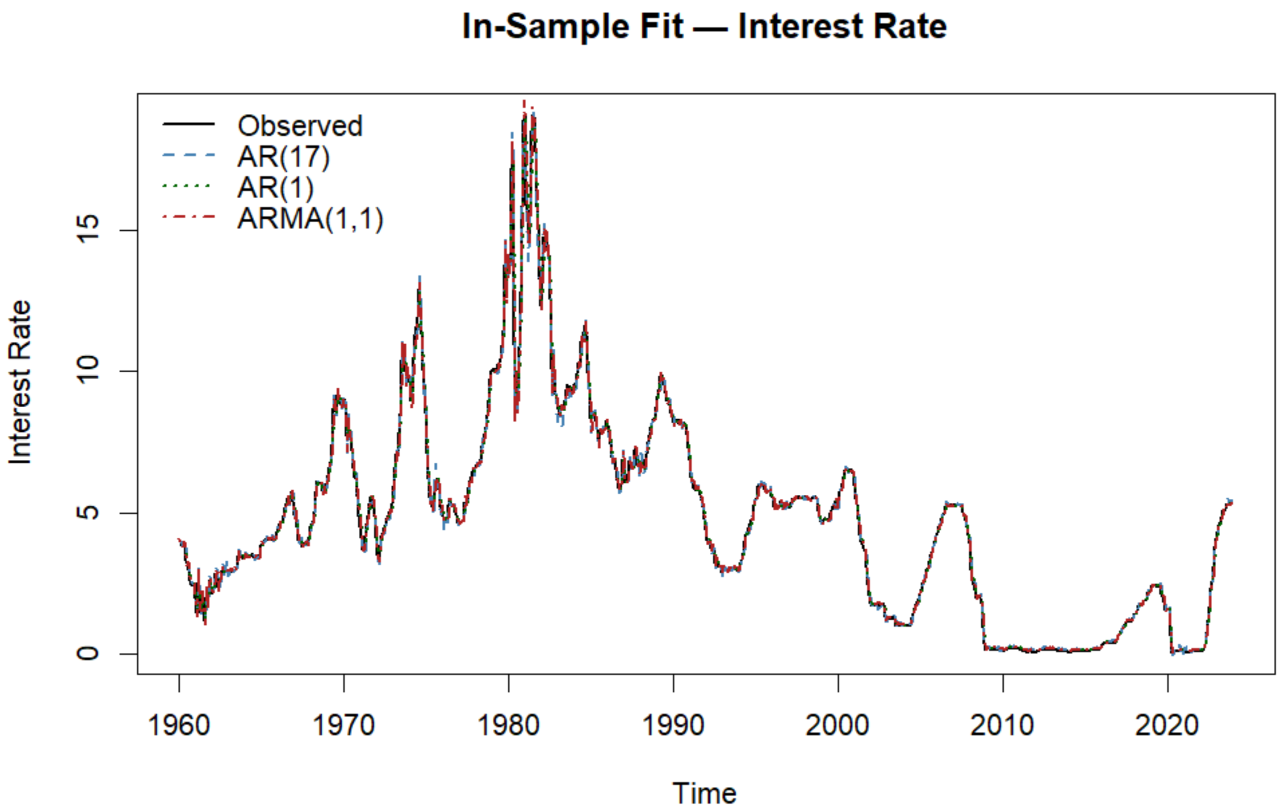

Figure 37: In sample fit: Federal Funds rate

Model	AIC	BIC	RMSE (train)	MAE (train)
AR(17)	903.55	991.78	0.4235	0.2224
AR(1)	1126.04	1139.97	0.5004	0.2436
ARMA(1,1)	980.55	999.12	0.4545	0.2246

Table 23: In-sample performance of candidate models for the Federal Funds rate

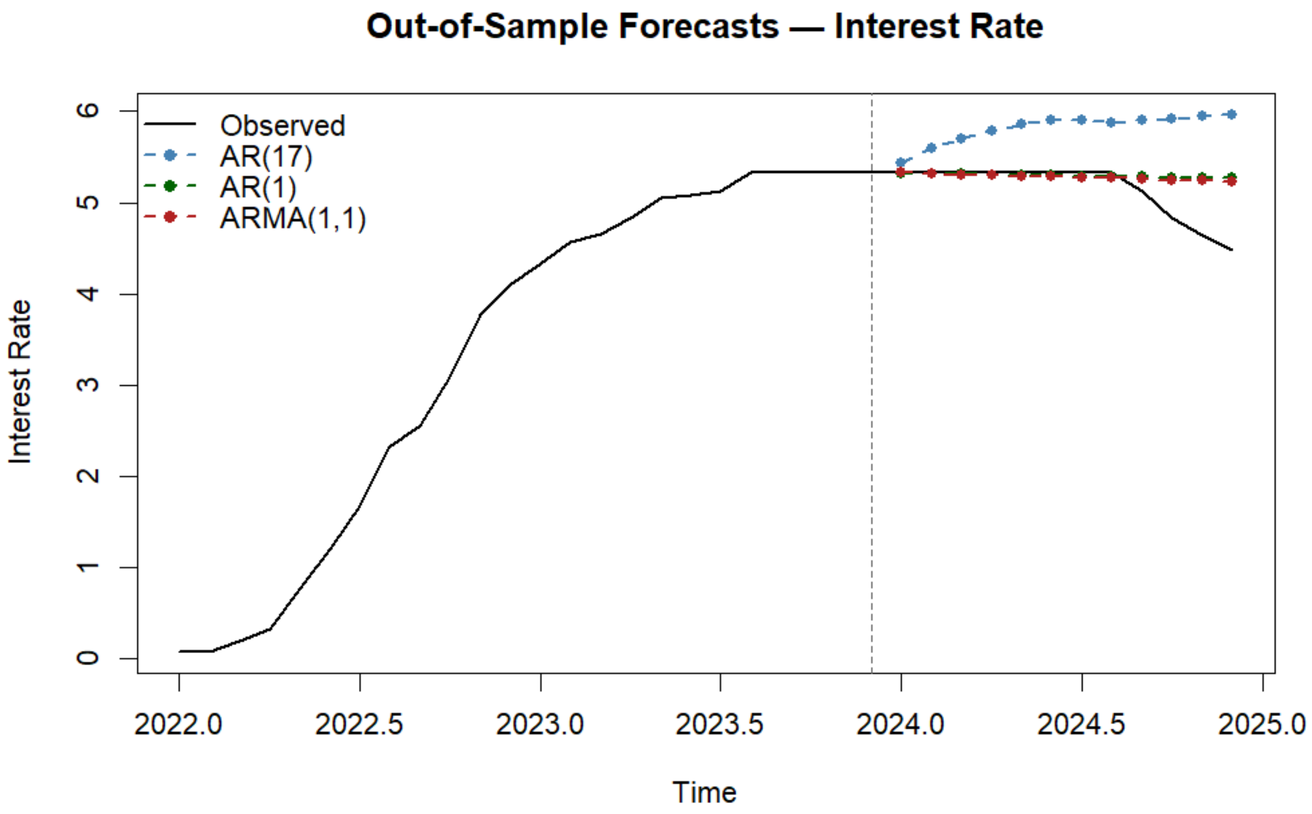

Figure 38: Out-of-sample forecasts: Federal Funds rate

Model	RMSE (test)	MAE (test)
AR(17)	0.7827	0.6727
AR(1)	0.3225	0.1844
ARMA(1,1)	0.3080	0.1815

Table 24: Out-of-sample forecast accuracy for the interest rate (h = 12)

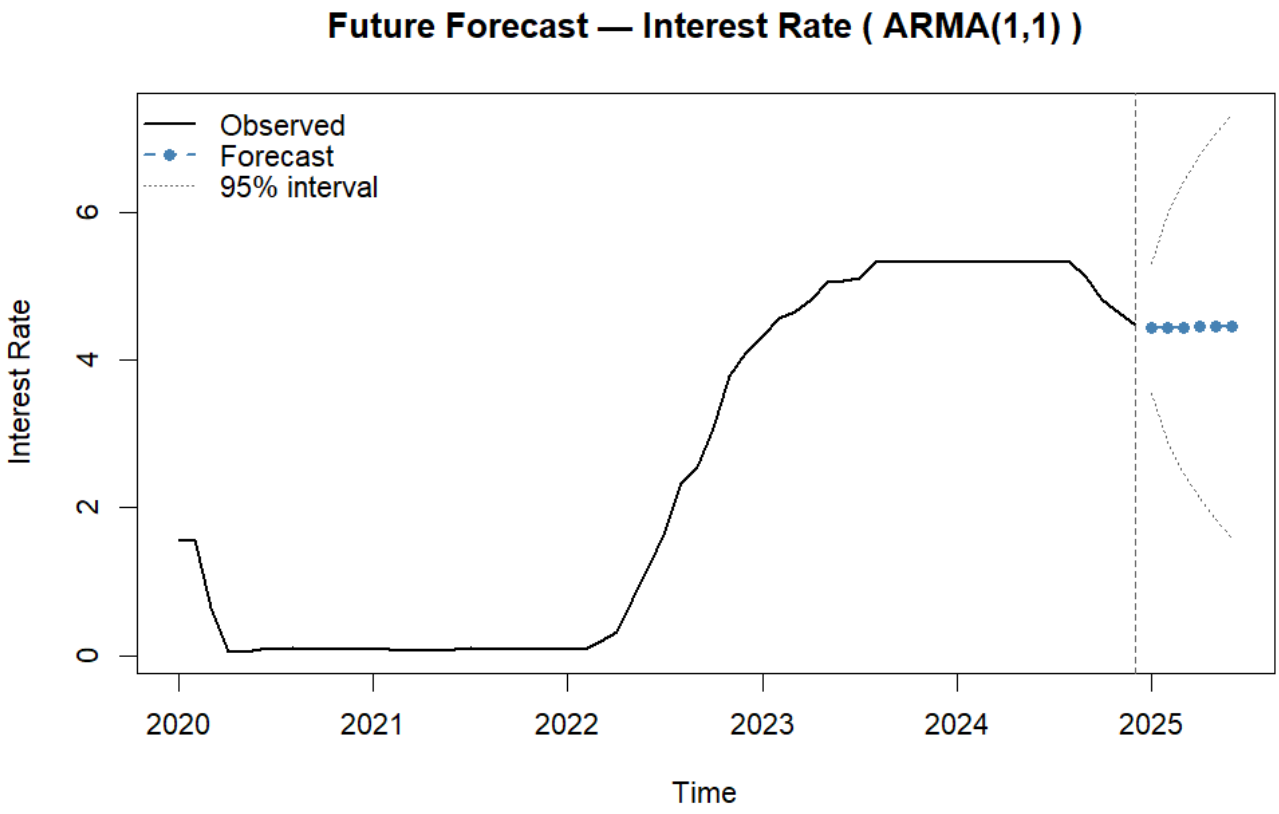

Figure 39: Future forecasts: Federal Funds rate

Period	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2025	4.4326	3.8533	5.0119	3.5467	5.3185
Feb 2025	4.4388	3.4310	5.4467	2.8975	5.9802
Mar 2025	4.4450	3.1515	5.7834	2.4669	6.4230
Apr 2025	4.4510	2.9318	5.9701	2.1277	6.7743
May 2025	4.4569	2.7478	6.1660	1.8430	7.0707
Jun 2025	4.4627	2.5882	6.3371	1.5960	7.3294

Table 25: Future forecasts for the Fedearal Funds rate using the ARMA(1,1) model

The training set of the Federal Funds rate series is from January 1960 to December 2023, we keep a test of 12 monthly observations which correspond to 1 year (2024). In addition, we choose to perform a 6 steps ahead future forecasts (above our observed data), which correspond to the first 6 months of 2025.

Figure 37 and Table 23 illustrate the results for the in-sample fit of the Federal Funds rate, namely r_t. We can observe that all the three models present quite similar results. Graphically, the trend of the series is well replicated by each models. However, in terms of performance, the AR(17) is the one with the lowest RMSE, MAE, AIC and BIC values, which indicate that this model is the best for in-sample fit, this is consistent with results obtained in question 1.

We made out-of-sample forecasts for the year 2024. We observe in Figure 38 that both ARMA(1,1) and AR(1) models predict well until the middle of 2024, but this is likely due to the fact that the trend was flat. In contrast, the AR(17) model did not predict well at any time. These results are highlighted in Table 24. Indeed, AR(17) have a RMSE of 0.7827, well above 0.3225 for AR(1) and 0.3080 for ARMA(1,1). These values indicate that AR(17) have poor out-of-sample performance.

As the ARMA(1,1) have the best out-of-sample accuracy we selected it for computing future forecasts on the first 6 months of 2025. We have the empirical mean of the series:

$$\bar{r} = \frac{1}{780}\sum_{t=1}^{780} r_t = 4.799372$$

Same as for g_t and π_t, the future forecasts of the r_t series are close to the empirical mean from January 2025 onward as it is illustrated in Figure 39 and in Table 25.

To sum up, even if the AR(17) provides the best in-sample performance, this model may suffer from overfitting in terms of out-of-sample forecasts, this highligts an important limitation of model selection based on in-sample criteria only. In particular, the test period (2024) corresponds to a change in the monetary policy regime. Indeed, this is the beginning of an easing cycle and therefore structural change in comparison to the training period. simpler models like AR(1) or ARMA(1,1) are more robust when structural breaks occur, while the AR(17) model, which captures dynamics of the historical data, may overfits patterns that are no longer the same in the new regime (in 2024).

3 Problem III

In this section, we have chosen to use Apple's stock returns from January 1, 2017, to December 30, 2025. To work with this data, we utilize the *quantmod* library in **R**.

3.1 Log-returns.

It is better to work with log-returns for several reasons.
First, log-returns help achieve stationarity. Prices generally increase over time and are not stationary, but stationarity is required for ARMA-GARCH models. Log-returns are stationary, allowing us to use these models correctly.

Second, they are easier to interpret because they represent continuous rates of return, rather than absolute price changes, making them comparable across time and assets.

Log-returns are also additive over time, so multi-period returns can be obtained by summing individual log-returns, which simplifies analysis and forecasting.

Moreover, they have an unconstrained range $(-\infty, +\infty)$ and a more symmetric distribution closer to normality than raw prices, which helps statistical modeling.

Finally, when comparing different assets, log-returns remove scale effects, allowing meaningful comparisons regardless of price level.

3.2 Descriptive statistics of Log Returns

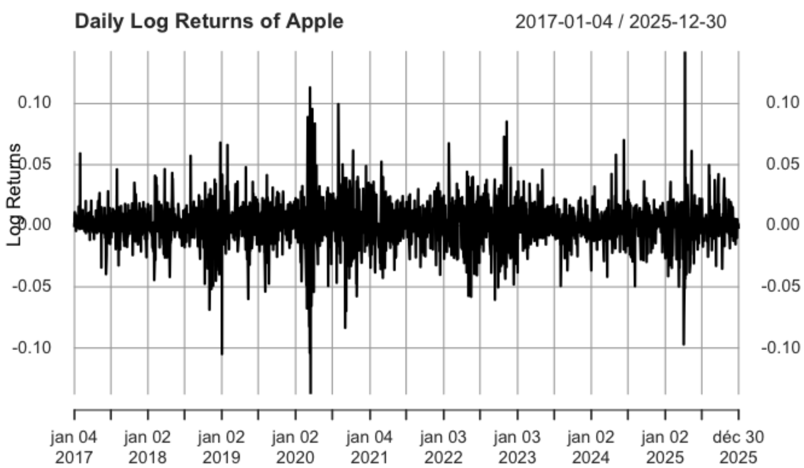

Figure 40: Daily of Apple log-returns

Apple's daily log-returns fluctuate around zero, consistent with the stationarity assumption for ARMA-GARCH models. Volatility clustering is evident, with high-volatility periods followed by calm periods, notably during early 2020 (COVID-19 crash) and around 2025. Overall, the series shows the typical stylized facts of financial returns, zero mean, volatility clustering, and occasional large spikes, justifying the use of a GARCH model.

Statistic	Value
Mean	0.001
Variance	3e-04
Skewness	-0.0824
Kurtosis	6.6484

Table 26: Summary statistics of Log Returns

The summary statistics in Table 1 highlight typical features of daily log-returns in financial markets. The mean (0.001) reflects very small daily price changes, while the variance (0.0003) indicates moderate volatility. Slightly negative skewness (-0.0824) suggests a small asymmetry toward losses, and high excess kurtosis (6.6484, well above 0 for a normal distribution) indicates fat tails, meaning extreme events occur more often than under a normal distribution. These characteristics emphasize the importance of accounting for rare but significant events when analyzing market risk.

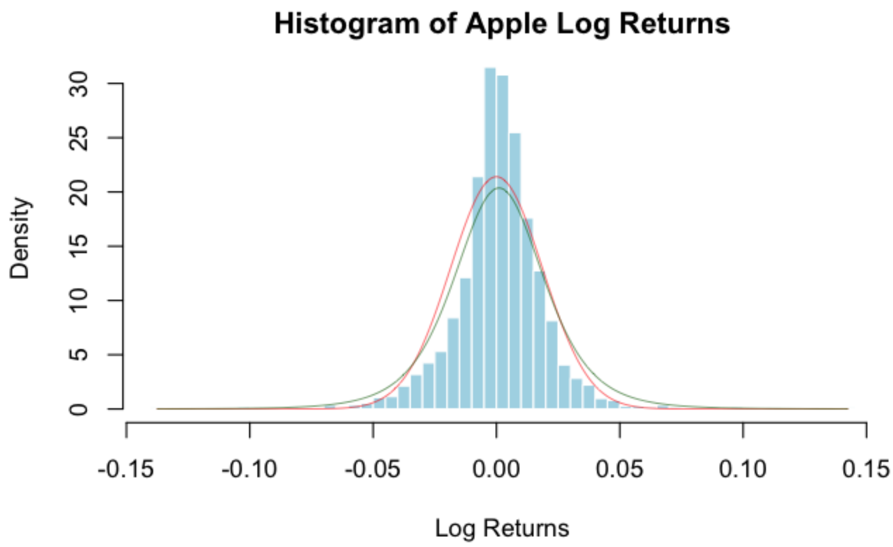

Figure 41: Histogram of Apple Log Returns with Normal (red) and Student (green) Densities

We observe in this histogram that the Student density approximates the Apple Log Returns better than the Normal density. Indeed, it has fatter tails, which explains that extreme values are more probable and the decay of the tails is more similar to the Student than to the Normal density. Therefore, when fitting the ARMA-GARCH model, using a Student distribution would better capture the distributional properties of our data compared to the Normal distribution.

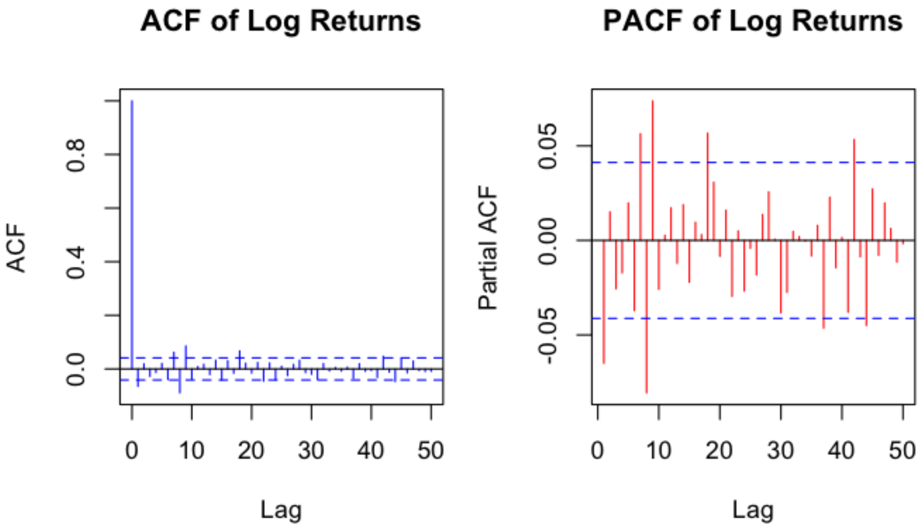

Figure 42: ACF and PACF of Apple Log Return

The ACF of Apple's daily log-returns shows no significant autocorrelation beyond lag 0, and the PACF has only small spikes within ± 0.07. This indicates that the series is essentially uncorrelated, consistent with financial theory that daily returns are nearly independent. Small deviations are negligible, but variability changes over time, suggesting volatility clustering that could be modeled with a GARCH process.

3.3 ARMA-GARCH model

To find the best ARMA-GARCH model, we proceeded in two steps. First, we tested all ARMA(p,q) combinations with $p, q \in \{0, \ldots, 3\}$ and selected the one with the lowest AIC, which penalizes models with too many parameters to avoid overfitting. We then applied the ARCH test of Engle on the residuals, which strongly rejected the null hypothesis of no ARCH effects (p-value ¡ 2.2e-16), confirming the need for a GARCH component. In a second step, we tested GARCH(p,q) models for $p, q \in \{0, \ldots, 2\}$with the ARMA orders fixed, and again selected the one with the lowest AIC. The final model was estimated using maximum likelihood with a Student distribution, consistent with the fat tails observed in our data.

For the ARMA, the lowest AIC is obtained for an ARMA(1,0) (AIC = -11591.26). For the GARCH, the lowest AIC is obtained for a GARCH(1,1) (AIC = -5.4449).

Table 27: Model selection based on AIC

Model	**AIC**
ARMA(1,0)	-11591.26
GARCH(1,1)	-5.4449
ARMA(1,0)-GARCH(1,1)	**-5.4487**

Table 28: Estimated coefficients of the ARMA(1,0)-GARCH(1,1) model

Parameter	**Estimate**	**Std. Error**	**t-value**	**p-value**
μ	0.001474	0.000281	5.251	0.000***
ar_1	0.006828	0.021065	0.324	0.746
ω	0.000010	0.000003	3.544	0.000***
α_1	0.111762	0.005810	19.238	0.000***
β_1	0.868237	0.012120	71.638	0.000***
ν	4.329723	0.260456	16.624	0.000***

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

When estimating the ARMA(1,0)-GARCH(1,1) model, we find that the AR(1) coefficient is not statistically significant (p-value = 0.746), suggesting that a pure GARCH(1,1) model may be sufficient to capture the dynamics of Apple log-returns. We therefore formally compare the two nested models.

Table 29: Model comparison: ARMA(0,0)-GARCH(1,1) vs ARMA(1,0)-GARCH(1,1)

Model	AIC	BIC	Log-likelihood
ARMA(0,0)-GARCH(1,1)	−5.4458	−5.4331	6158.730
ARMA(1,0)-GARCH(1,1)	−5.4449	−5.4297	6158.783

Student-t distribution. Lower AIC/BIC indicates preferred model.

Table 29 reports the information criteria and log-likelihoods for both specifications. The ARMA(0,0)-GARCH(1,1) achieves a lower AIC (−5.4458 vs −5.4449) and a lower BIC (−5.4331 vs −5.4297) than the ARMA(1,0)-GARCH(1,1), while the log-likelihoods are virtually identical ($\Delta = 0.05$). The BIC difference is particularly telling, as BIC penalises additional parameters more heavily than AIC, the extra AR(1) term therefore costs more than it contributes. We conclude that the mean dynamics of Apple log-returns are fully captured by a constant, and retain the ARMA(0,0)-GARCH(1,1) as our final model for the remainder of the analysis.

3.4 Diagnostics

To check if the model is well specified, we perform a Box-Pierce test on the standardized residuals to verify that no autocorrelation remains, and on the squared standardized residuals to check that no ARCH effects remain. In both cases, the null hypothesis is that the residuals are white noise.

Table 30: Diagnostic Tests on Standardized Residuals

Test	Statistic	df	p-value	Conclusion
Box-Pierce (residuals)	10.749	10	0.3774	No autocorrelation
Box-Pierce (squared residuals)	7.5477	10	0.6729	No ARCH effects

The Box-Pierce tests on both standardized residuals and squared standardized residuals fail to reject the null hypothesis (p-values of 0.38 and 0.67 respectively), indicating that the residuals behave as white noise and that no ARCH effects remain. This confirms that the ARMA(0,0)-GARCH(1,1) model is well specified and successfully captures both the mean and variance dynamics of Apple log-returns.

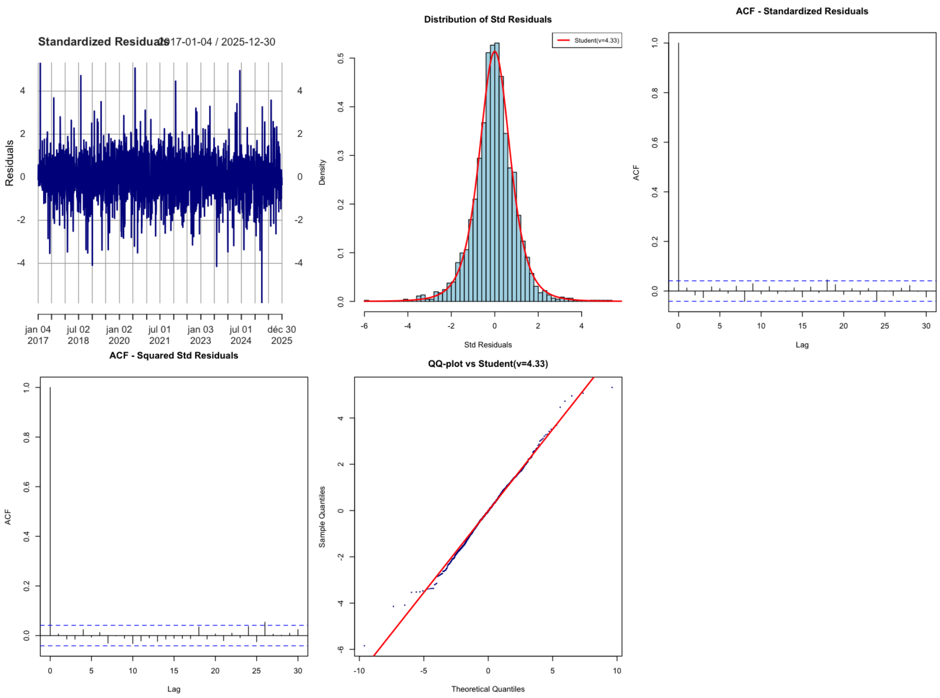

Figure 43: Residual diagnostics for the ARMA(0,0)-GARCH(1,1) model.

Figure 43 presents the full diagnostic output for the ARMA(0,0)-GARCH(1,1) model estimated with a Student-t distribution.

Standardized residuals: The residual series fluctuates around zero with no visible trend or structure, and the large spikes are isolated rather than clustered, suggesting that the GARCH(1,1) component has successfully absorbed the volatility clustering present in the raw returns.

ACF of standardized residuals: All autocorrelations lie within the 95% confidence bands across all 30 lags, with the exception of one marginally significant spike around lag 20. This is consistent with the Box-Pierce test result ($p = 0.39$) and confirms the absence of remaining serial correlation in the mean equation.

ACF of squared standardized residuals: No significant spike is visible at any lag, confirming that all ARCH effects have been captured by the GARCH(1,1) variance equation. This is corroborated by the

Box-Pierce test on squared residuals ($p = 0.67$).

Distribution of standardized residuals: The histogram aligns closely with the fitted Student($\hat{\nu} = 4.33$) density, confirming that the distributional assumption is appropriate. The estimated degrees-of-freedom parameter $\hat{\nu} = 4.33$ is highly significant and well below 30, confirming the presence of heavy tails that a Gaussian assumption would severely understate.

QQ-plot against Student($\hat{\nu} = 4.33$). The bulk of the distribution aligns well along the 45-degree line, with only a handful of extreme observations in the left tail deviating slightly. These few outliers correspond to exceptional market events such as the COVID-19 crash of March 2020, which no parametric model can fully internalize. Overall, the Student-t assumption provides a good fit for the tail behaviour of Apple log-returns.

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